



Republic of Iraq
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Computer Network Fundamentals

Chapter One

Introduction to Networking

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Introduction to Networking

A network is a set of devices (nodes) connected by communication links; each node is capable of sending and/or receiving data generated by other nodes on the network.

1.1- Data Communication:

- ❖ When we communicate, we are sharing information. This sharing can be **local** or **remote**. *Local communication* usually occurs face to face, while *remote communication* takes place over distance.
- ❖ The term *Telecommunication*, which includes telephony and television, means *communication at a distance*.
- ❖ *Data communication* is the exchange of *data* (in the form of 0's and 1's) between two *devices* via some form of *transmission medium* (such as a wire cable).
- ❖ The effectiveness of a data communications system depends on three fundamental characteristics: (Delivery, Accuracy, and Timeliness)
 1. **Delivery:** The system must deliver data to the correct destination; Data must be received by the intended device or user.
 2. **Accuracy:** The system must deliver the data accurately; Data that have been altered (delay) in transmission and left uncorrected are unusable.
 3. **Timeliness:** The system must deliver data in a timely manner; Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called *real-time* transmission.

1.2- Components:

A data communication system is made up of five components (see Figure 1).

1. **Message**: The message is the information (data) to be communicated. It can consist of text, numbers, pictures, sound, or video or any combination of these.
2. **Sender**: The sender is the device that sends the data message. It can be a **computer**, workstation, mobile phone, video camera, and so on.
3. **Receiver**: The receiver is the device that receives the message. It can be a computer, workstation, mobile phone, television, and so on.
4. **Medium**: The transmission medium is the physical path by which a message travels from-sender to receiver. It can consist of twisted pair wire, coaxial cable, fiber- optic cable, laser, or radio waves.
5. **Protocol**: A protocol is a set of rules that govern data communication. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating; just as a person who speaks, only Japanese cannot understand a person speaking French.

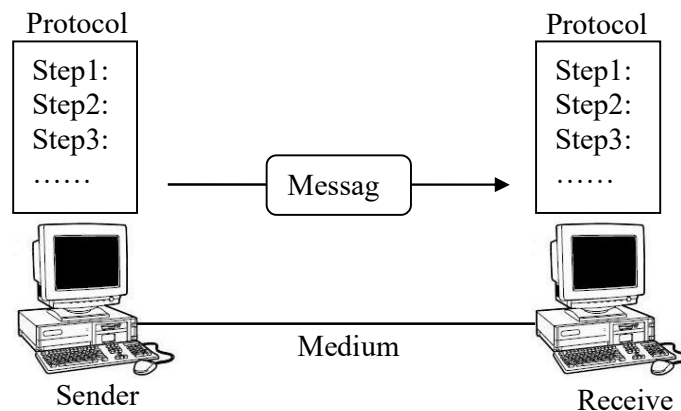


Figure (1) Components of a Data Communication System

1.3 Network Criteria:

To be considered effective and efficient, a network must meet a number of criteria. The most important of these are *performance*, *reliability*, and *security*.

(A) Performance:

Performance can be measured in many ways, including *transit time* and *response time*.

Transit time is the amount of time required for a message to travel from one device to another.

Response time is the elapsed time between an inquiry and a response.

The performance of a network *depends on* a number of factors:

✓ *Number of users:*

Having a large number of concurrent users can slow response time in a network not designed to coordinate heavy traffic loads. The design of a given network is based on an assessment of the average number of users that will be communicating at any one time. How a network responds to loading is a measure of its performance.

✓ *Type of transmission medium:*

The medium defines the speed at which data can travel through a connection (the data rate). Today's networks are moving to faster and faster transmission media, such as fiber optic cabling. A medium that can carry data at 1000 megabits per second is ten times more powerful than a medium that can carry data at only 100 megabits per second. Today transmission speeds have increased to 10Gps and 40Gps

(B) Reliability:

In addition to accuracy of delivery, network reliability is measured by frequency of failure, the time it takes a link to recover from a failure, and the network's robustness.

✓ *Frequency of failure:*

All networks fail occasionally. A network that fails often, however, is of little value to a user.

✓ ***Recovery time of a network after a failure:***

How long does it take to restore service? A network that recovers quickly is more useful than one that does not.

✓ ***Catastrophe:***

Networks must be protected from catastrophic events such as fire or theft. One protection against unforeseen damage is a reliable system to back up network software.

(C) Security:

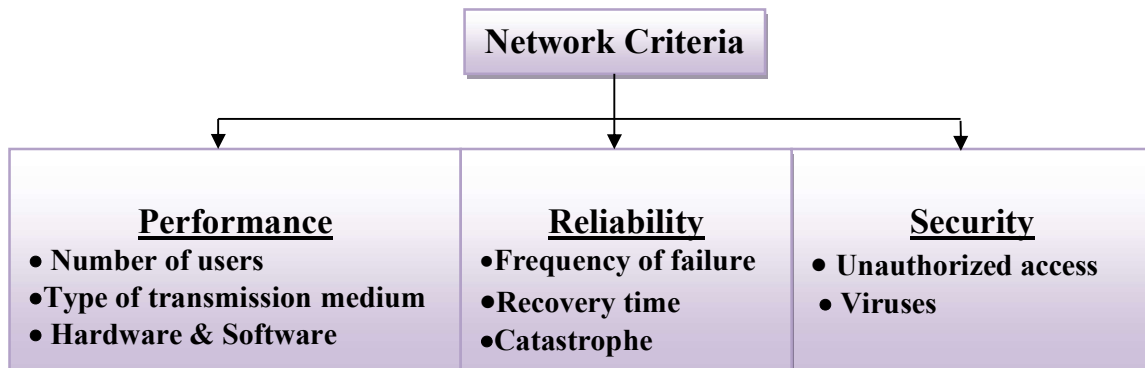
Network security issues include protecting data from unauthorized access and viruses.

✓ ***Unauthorized access:***

For a network to be useful, sensitive data must be protected from unauthorized access. Protection can be accomplished at a number of levels. At the lowest level are user identification codes and passwords. At a higher level are encryption techniques.

✓ ***Viruses:***

Because a network is accessible from many points, it can be susceptible to computer viruses. A virus is an illicitly introduced code that damages the system. A good network is protected from viruses by hardware and software designed specifically for that purpose.



1.4- Basic Concepts of Networking

Before examining the specifics of how data are transmitted from one device to another it is important to understand the relationship between the communicating devices. Five general concepts provide the basis for this relationship.

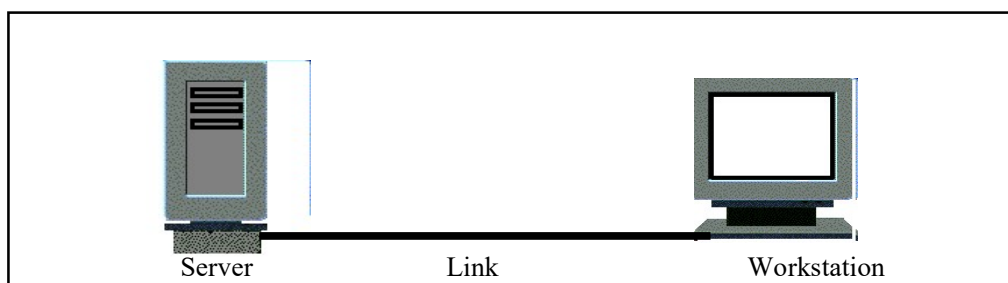
- Line configuration.
- Topology.
- Transmission mode.
- Categories of networks.
- Internetworks.

1.4.1- Line Configuration

A link is the physical communication pathway that transfers data from one device to another. Line configuration defines the attachment of communication devices to a link. There are two possible line configurations:

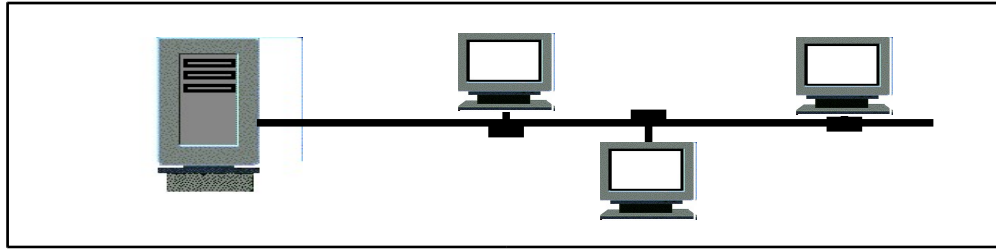
Point-to-Point

A point-to point line configuration provides a **dedicated** link between two devices. The entire capacity of the channel is reserved for transmission between those two devices. Most point-to-point line configurations use an actual length of wire or cable to connect the two ends but other options, such as microwave or satellite links are also possible.



Multipoint

A multipoint (also called multidrop) line configuration is one in which more than two specific devices share a single link. In a multipoint environment, the capacity of the channel is **shared**, either spatially or temporally. If several devices can use the link simultaneously, it is a spatially shared line configuration. If users must take turns, it is a time-shared line configuration.



1.4.2- Topology

The term topology refers to the way a network is laid out, either *physically* or *logically*. Two or more devices connect to a link; two or more links form a topology. ***The topology of a network is the geometric representation of the relationship of all the links and linking devices (nodes) to each other.***

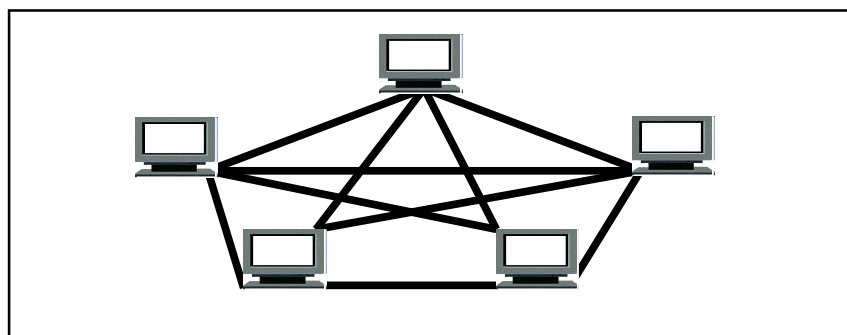
There are five basic topologies possible. These five labels describe how the devices in a network are interconnected rather than their physical arrangement.

Two relationships are possible: ***peer-to-peer***, where the devices, share the link equally, and ***client-server***, where one device controls traffic and the others must transmit through it. Ring and mesh topologies are more convenient for peer-to-peer transmission, while star and tree are more convenient for client-server. A bus topology is equally convenient for either.

❖ Mesh Topology

In a mesh topology, every device has a dedicated point-to-point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects.

A fully connected mesh network therefore has $[n (n-1) / 2]$ physical channels to link it devices. (See Figure below).



A mesh offers **several advantages** over other network topologies:

1. The use of dedicated links guarantees that each connection can carry its data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
2. A mesh topology is robust; if one link becomes unusable, it does not affect the entire system.
3. Security where every message sent travels along a dedicated line, only the intended recipient sees it.
4. Point-to-point links make fault identification and fault isolation easy.

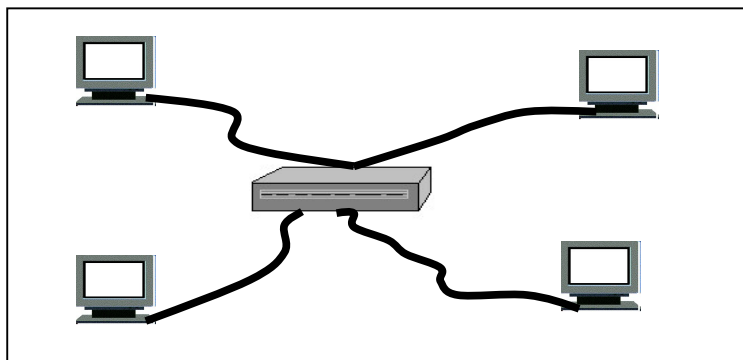
The main **disadvantages** of a mesh topology are:

1. Difficult installation and reconfiguration because every device must be connected to every other device.
2. The bulk of the wiring can be greater than the available space can accommodate.
3. The hardware required to connect each link can be expensive.

Due to these reasons, a mesh topology is usually of limited use

❖ **Star Topology**

In a star topology, each device has a dedicated point-to-point link only to a *central controller*, usually called a **hub or switch**. The devices are not linked to each other. Unlike a mesh topology, a star topology **does not** allow direct traffic between devices. The controller acts as an exchange. If one device wants to send data to another, it sends to the controller, which then relays the data to the other connected devices.



The **advantages** of star topology are:

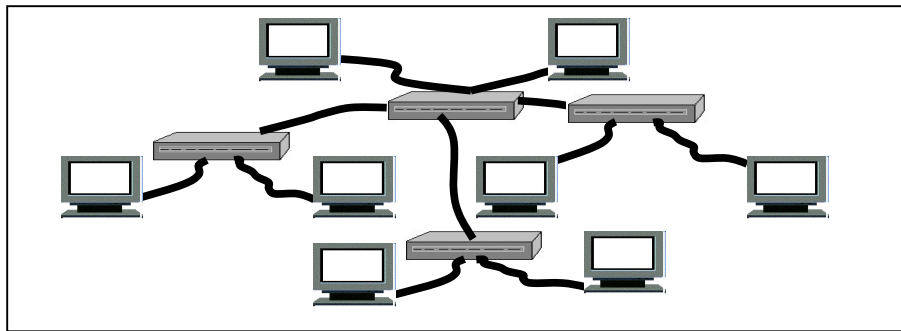
1. Less expensive than a mesh topology.
2. Easy to install.
3. Robustness; If one link fails, only that link is affected.
4. Easy fault identification and fault isolation; As long as the hub is working, it can be used to monitor link problems and bypass defective links.

The **disadvantages** of the star topology are:

1. Any fault occur to the central controller will drop down the entire network.
2. Although the star needs less cabling than mesh, it still needs more cables than ring and tree topology.

❖ **Tree Topology**

A tree topology is a variation of a star. However, not every device plugs directly into the central hub. The majority of devices connect to a secondary hub that in turn is connected to the central hub.



The central hub in the tree is an active hub. An **active hub** contains a repeater, which is a hardware device that regenerates the received bit patterns before sending them out. Repeating strengthens transmissions and increases the distance a signal can travel between sender and receiver.

The secondary hubs may be active or **passive hubs**. A passive hub provides a simple physical connection between the attached devices. Internally, each passive hub contains a set of resistors to balance the circuit linking the connected devices.

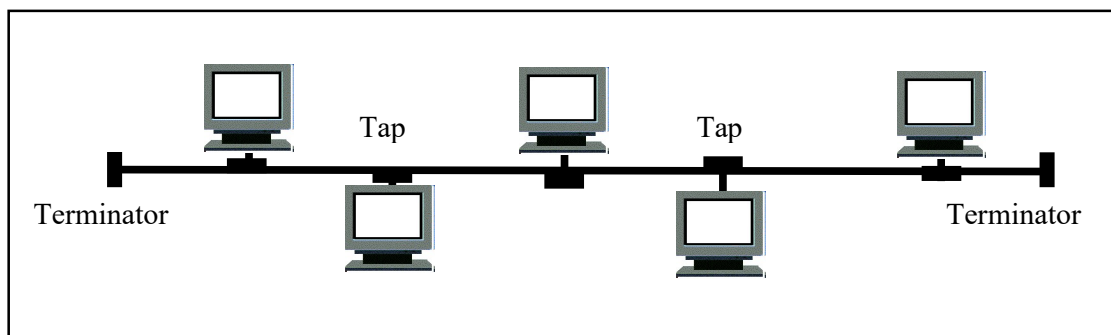
The advantages and disadvantages of a tree topology are generally the same as those of a star. The addition of secondary hubs however, brings two **further advantages**.

First, it allows more devices to be attached to a single central hub and can therefore increase the distance a signal can travel between devices.

Second, it allows the network to isolate and prioritize communications from different computers.

❖ Bus Topology

The preceding examples all describe point-to-point configurations. A bus topology on the other hand, is multipoint. One long cable acts as a backbone to link all the devices in the network.



Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection that runs between the device and the main cable, a tap is a connector that splices into the main. As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker the farther it has to travel. For this reason, there is a limit on the number of taps a bus can support and on the distance between those taps.

The advantages of a bus topology are:

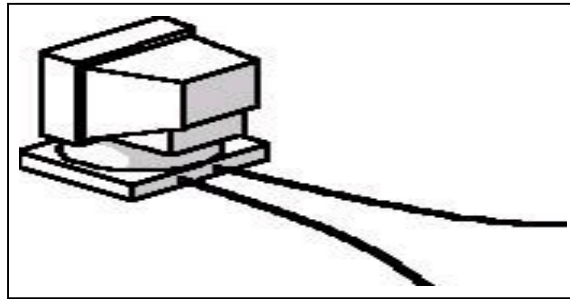
1. Ease of installation.
2. Use less cabling than mesh, star, and tree topologies.

While the disadvantages are:

1. Difficult reconfiguration and fault isolation.
2. A fault or break in the bus cable stops all transmission, even between devices on the same side of the problem.
3. Adding new devices may therefore require modification or replacement of the backbone.

❖ Ring Topology

In a ring topology, each device has a dedicated point-to-point line configuration only with the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination. Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along.



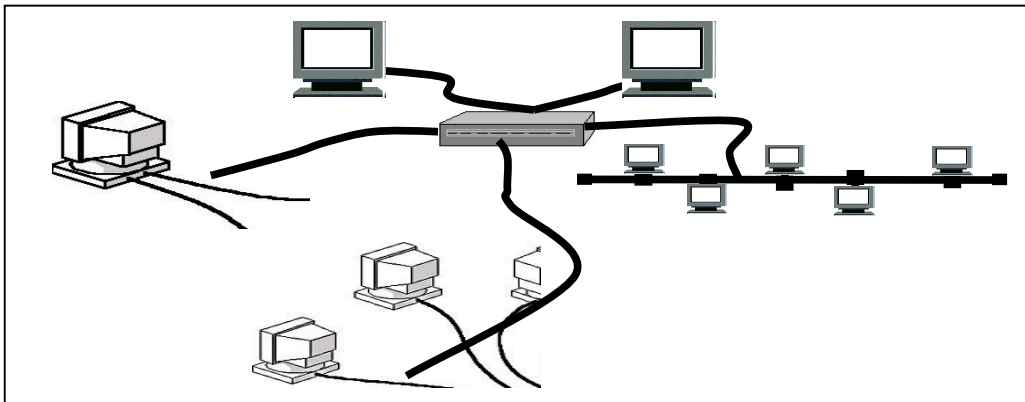
The advantages of ring topology are:

1. Relatively easy to install and reconfigure.
2. Addition or deletion of a device requires moving only two connections.

The disadvantage of the ring topology is that a break in the ring (such as a disabled station) can disable the entire network, and it is one way data flow.

❖ Hybrid Topologies

Often a network combines several topologies as sub-networks linked together in a larger topology. For instance, one department of a business may have decided to use a bus topology while another department has a ring. The two can be connected to each other via a central controller in a star topology.



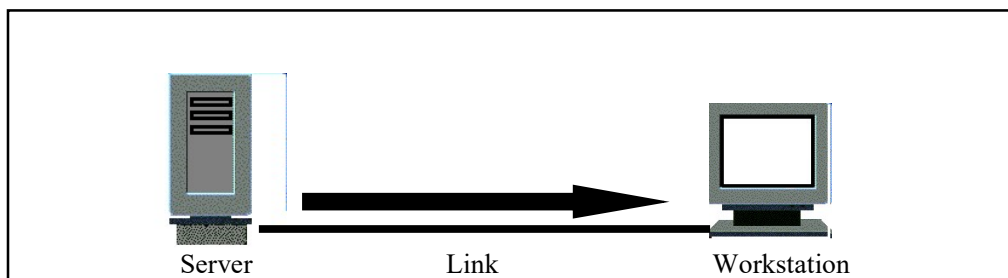
1.4.3- Transmission Modes

The term transmission mode is used to define the direction of signal flow between two linked devices. There are *three* types of transmission modes:

- ❖ *Simplex.*
- ❖ *Half duplex.*
- ❖ *Full duplex*

Simplex

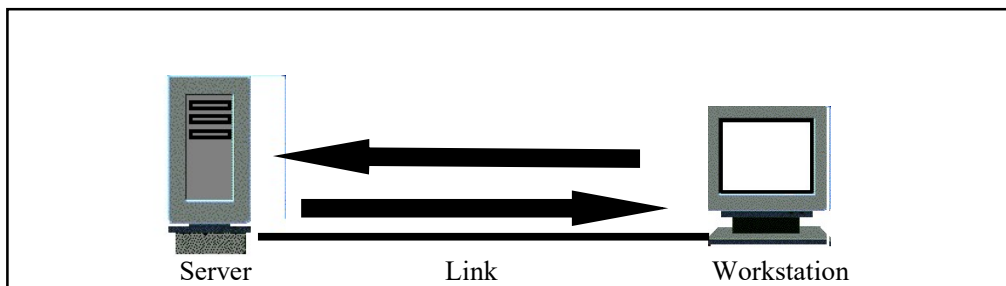
In, simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two stations on a link can transmit the other can only receive.



Keyboards and traditional monitors are both examples of simplex devices. The keyboard can only introduce input; the monitor can only accept output.

Half-Duplex

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa (see Figure below).

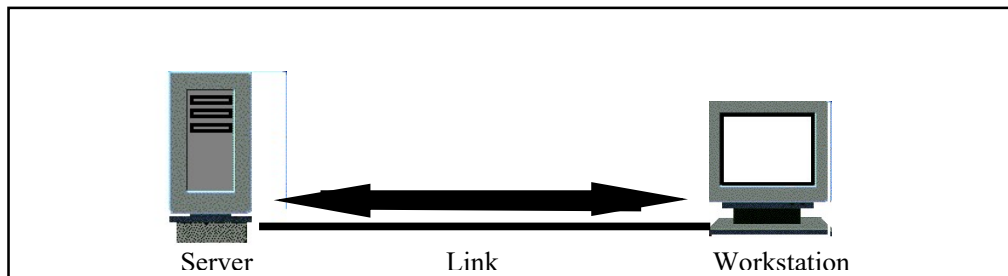


versa (see Figure below).

The half-duplex mode is like a one-lane road with two-directional traffic. In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time. Walkie-talkies are half-duplex systems.

Full-Duplex

In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously.



The full-duplex mode is like a two-way street with traffic flow in both directions at the same time. In full-duplex mode, signals going in either direction share the capacity of the link.

1.4.4- Categories of Networks

The category of a network can be determined

- According to its size.
- According to its physical architecture (center of control).
- According to transmission technology.
- According to its ownership.
- According to service providing.

(A) According to the size:

We are generally referring to three primary categories:

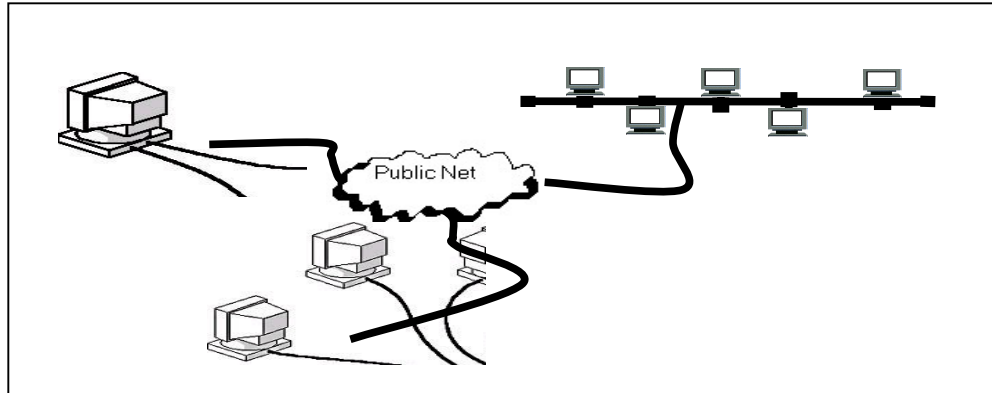
- ❖ *Local area networks (LANs).*
- ❖ *Metropolitan area networks (MANs).*
- ❖ *Wide area networks (WANs).*

❖ **Local Area Network (LAN)**

A local area network is usually privately owned and links the devices in a single office building, or campus. Depending on the needs of an organization and the type of technology used. A LAN can be as simple as two PCs and a printer in someone's home office or it can extend throughout a company and include voice, sound, and video peripherals. Currently, LAN size *is limited to a few Kilometers*.

❖ Metropolitan Area Network (MAN)

A metropolitan area network is designed to *extend over* an entire city. It may be a single network such as a cable television network or it may be a means of connecting a number of LANs into a larger network so that resources may be shared LAN-to-LAN as well as device-to-device. For example, a company can use a MAN to connect the LANs in all of its offices throughout a city.



A MAN may be wholly owned and operated by a private company, or it may be a service provided by a public company.

❖ Wide Area Network (WAN)

A wide area network provides *long-distance* transmission of data; voice, image, and video information over large geographical areas that may comprise a country, a continent, or even the whole world. In contrast to LANs (which depend on their own hardware for transmission), WANs may utilize public, leased, or private communication devices, usually in combinations, and can therefore span an unlimited number of miles. A WAN that is wholly owned and used by a single company is often referred to as an *enterprise network*.

(B) According to the physical architecture (center of control)

❖ Centralized Networks

A central computer "Mini computer or Mainframe" that manages all communication and requests in the network.

❖ **Distributed Network**

A group of Personal Computers (PC's) works together and share the same input / output devices.

❖ **Hybrid Network**

A combination of centralized and distributed networks

(C) According to the ownership

❖ **Public Network**

On a network that is owned, managed, and operated by a public company.

❖ **Private Network**

Owned and managed by a private organization

(D) According to transmission technology

❖ **Broadcasting Network**

It does not restrict or determine a specified destination. It has a single communication channel that is shared by all the machines on the network. Short messages, called packets sent by any machine are received by all the others. Broadcast systems generally also allow the possibility of addressing a packet to all destinations by using a special code in the address field.

❖ **Point to Point Network**

Data or packets are sent to a specify destination which is able to reflect an each of the receiving.

(E) According to service providing

❖ **Peer to Peer Network**

Any PC connected to this network can provide services to any other PC and request for services from any.

❖ **Client/Server Network**

The most popular network, the network depends on a PC act as a service Provider called the **SERVER**. The server restricts the same policy to control the determination of the client how will get the service and type of service.



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Computer Network Fundamentals

Chapter Two

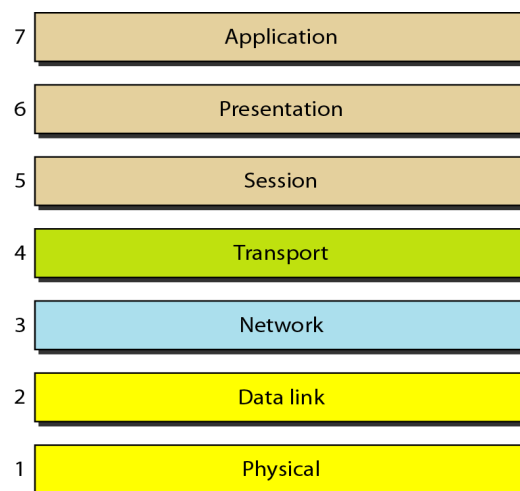
The OSI Model

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2.1- The OSI Model

The International Standards Organization (**ISO**) covers all aspects of network communications is the **Open Systems Interconnection (OSI)** model. An **open system** is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture. The **purpose of** the OSI model is to open communication between different systems without requiring, changes to the logic of the underlying hardware and software. The OSI is not a protocol, it is a model for understanding and designing a network architecture.

The OSI model consists of *seven* separate but related layers, each of which defines a segment of the process of moving information across a network:



2.1.1- OSI Layered Architecture:

Each layer defines a family of functions distinct from those of the other layers. Figure (2.1) shows the layers involved when a message sent from device **A** to device **B**. As the message travels from A to B, it may pass through many intermediate nodes. The intermediate nodes usually involve only the first three layers of the OSI model.

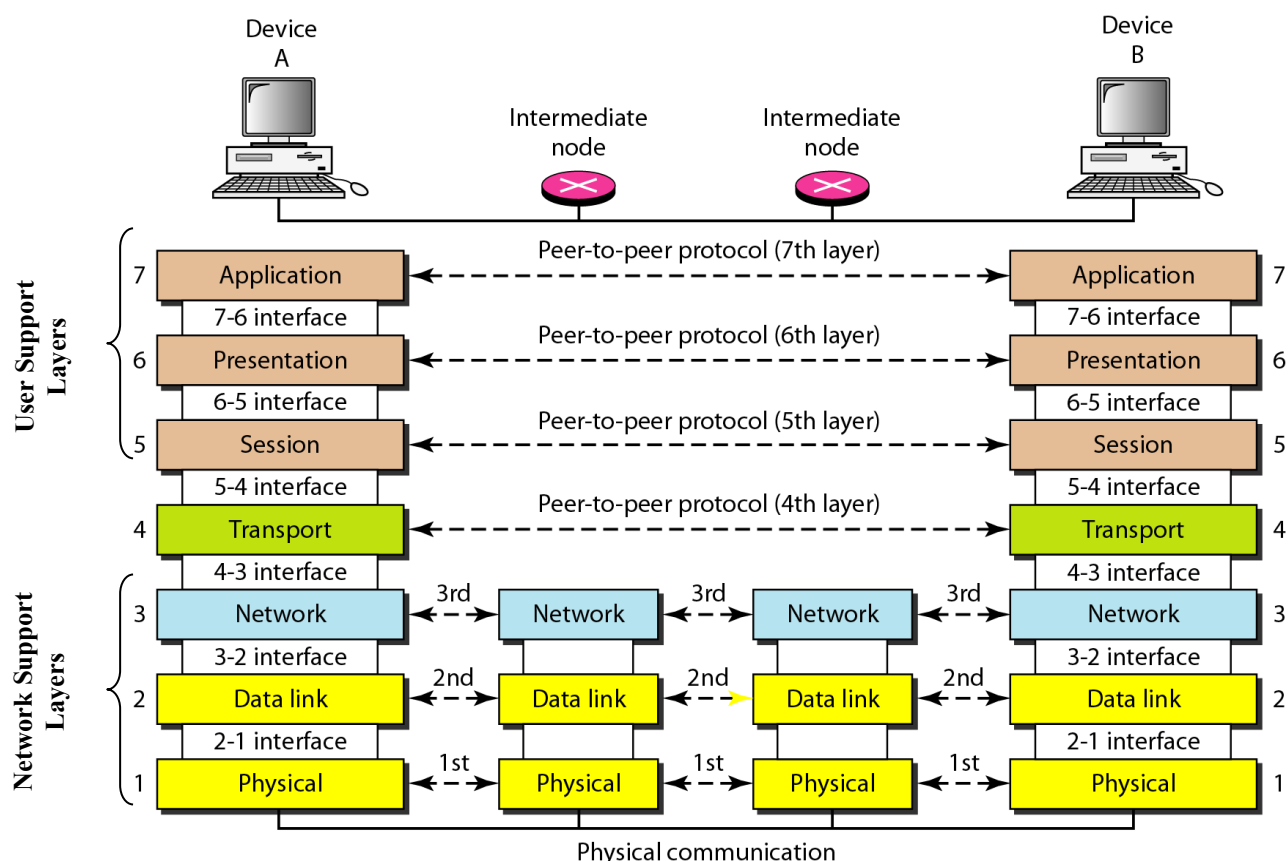


Figure (2.1) shows the layers involved when a message sent from device A to device B.

2.1.2- OSI Interfaces and peer-to-peer between Layers

Each interface defines what information and services a layer must provide for the layer above it, *Between layers in the same machine*. Layer 3, for example, uses the services provided by layer 2 and provides services for layer 1

Between machines, layer x on one machine communicates with layer x on another machine. This communication is governed by an agreed upon series of rules and conventions called **protocol**. Each layer in the sending machine **adds** its own information to the message it receives from the layer just above it, and **passes** the whole package to the layer just below it. This information added in the form of **headers** or **trailers** (control data appended to the beginning or end of a data parcel).

2.1.3- Encapsulation and Protocol data unit (PDU)

As application data is passed down the protocol stack on its way to be transmitted across the network media, various protocols add information to it at each level. This is commonly known as the encapsulation process.

The form that a piece of data takes at any layer is called a Protocol Data Unit (PDU). During encapsulation, each succeeding layer encapsulates the PDU that it receives from the layer above in accordance with the protocol being used. At each stage of the process, a PDU has a different name to reflect its new functions.

- **Data** - The general term for the PDU used at the application layer
- **Segment** - Transport layer PDU
- **Packet** - Network layer PDU
- **Frame** - Data Link layer PDU
- **Bits** - A PDU used when physically transmitting data over the medium

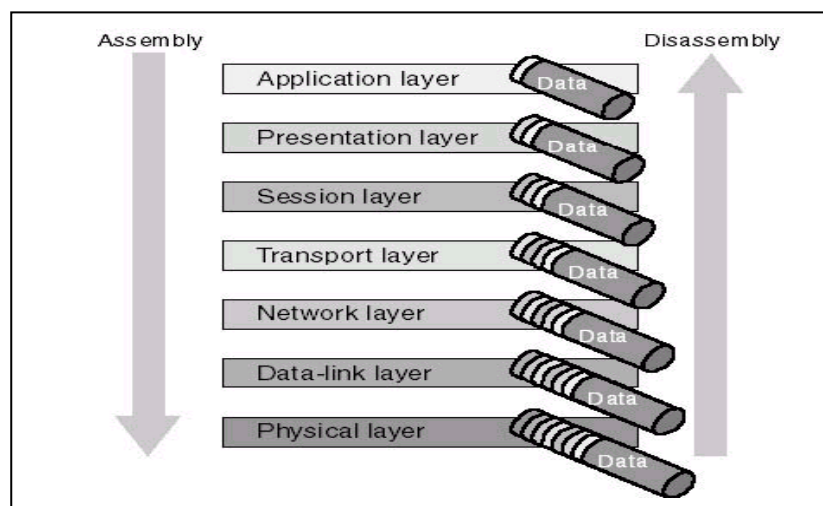


Figure (2.2): The encapsulation process

Figure (2.2), Demonstrates the encapsulation process. The process starts at layer 7 (the application layer), then moves from layer to layer in descending, sequential order. At each layer, a header, or possibly a trailer, can be added to the data unit. Commonly, the trailer is added only at layer 2. When the formatted data unit passes through the physical layer (layer 1), it is changed into an electromagnetic signal and transported along a physical link.

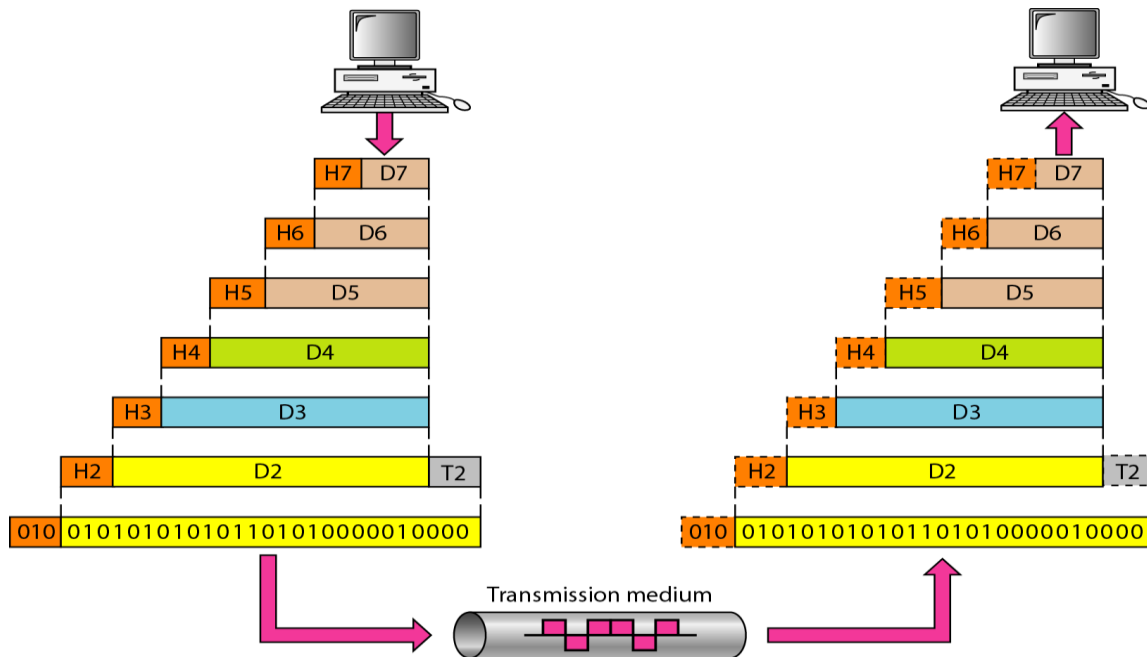


Figure (2.3): An exchange using the OSI model

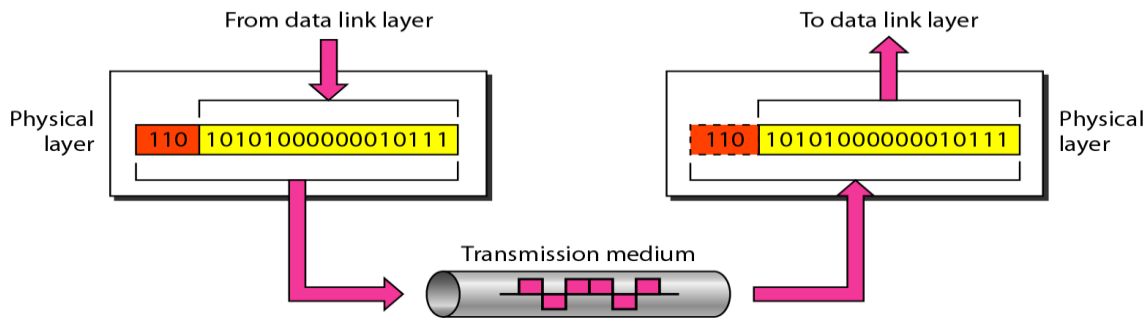
Upon reaching its destination, the signal passes into layer 1 and is transformed back into digital form. The data units then move back up through the OSI layers. As each block of data reaches the next higher layer, the headers and trailers attached to it at the corresponding sending layer are removed, and actions appropriate to that layer are taken. By the time it reaches layer 7, the message is again in a form appropriate to the application and is made available to the recipient.

2.2- Functions of the Layers

2.2.1- Physical Layer

The physical layer coordinates the functions required to transmit a bit stream over a physical medium. It deals with the mechanical and electrical specifications of the primary connections, such as cables, connectors, and signalling options that physically link two nodes on a network.

This first layer receives a data unit from the second layer and puts it into a format capable of being carried by a communications link. It oversees the changing of a bit streams into electromagnetic signals, and their transmission onto and across a medium.

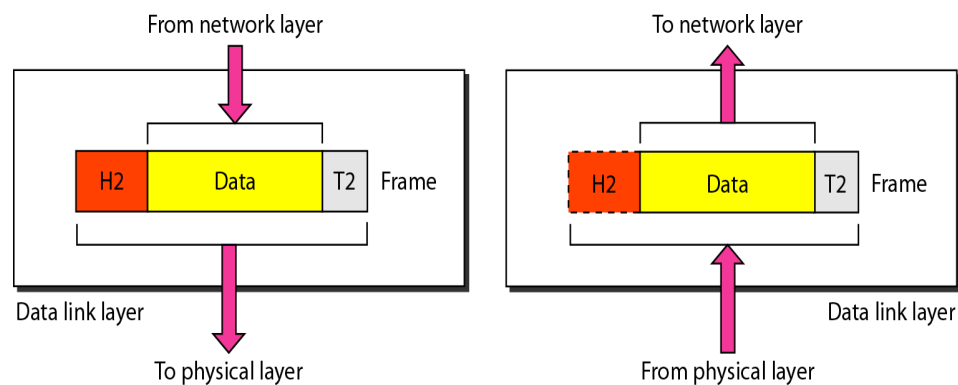


This task requires a number of considerations:

- ❖ **Line configuration:** How can two or more devices are linked physically? Are transmission lines to be shared or limited to use between two devices? Is the line available or not?
- ❖ **Data transmission mode:** It does transmission flow one-way or both ways between two connected devices. Or does it alternate?
- ❖ **Topology:** How network devices arranged? Do they pass data directly to each other or through an intermediary? And by what paths?
- ❖ **Signals:** What type of signals is useful for transmitting information?
- ❖ **Encoding:** How are bits (0s and 1s) to be represented by available signalling systems? How data are represented by signals?
- ❖ **Medium:** What is the physical environment for the transmission of data?
- ❖ **Multiplexing:** Using a single physical line to carry data between many devices at the same time.

2.2.2- Data link Layer

The data link layer is **responsible for delivering data units (groups of bits) from one station to the next without errors**. It accepts a data unit from the third layer and **adds** meaningful bits to the beginning (header) and end (trailer) that contain addresses and of control information. A data unit with this additional information called a **Frame**.



The responsibilities of the data link layer include the following:

- ❖ **Node-to-node delivery:** The data link layer is responsible for node-to-node delivery.
- ❖ **Physical Addressing:** Headers and trailers added at this layer include the *physical address* of the most recent node and the next intended node.
- ❖ **Access control:** When two or more devices connected to the same link, the data-link layer protocols are necessary to determine which device has control over the line at any given time.
- ❖ **Flow control:** To avoid overwhelming the receiver, the data link layer regulates the amount of data that can be transmitted at one time. It adds identifying numbers to enable the receiving node to control the ordering of the frames.
- ❖ **Error handling:** Data link layer protocols provide for data recovery, usually by having the entire frame retransmitted.

2.2.3- Network Layer

The network layer is responsible for the source-to-destination delivery of a packet across multiple network links. Whereas the data link layer oversees *node-to-node* delivery, the network layer **ensures** that each **packet** gets from its point of origin to its final destination successfully and efficiently.

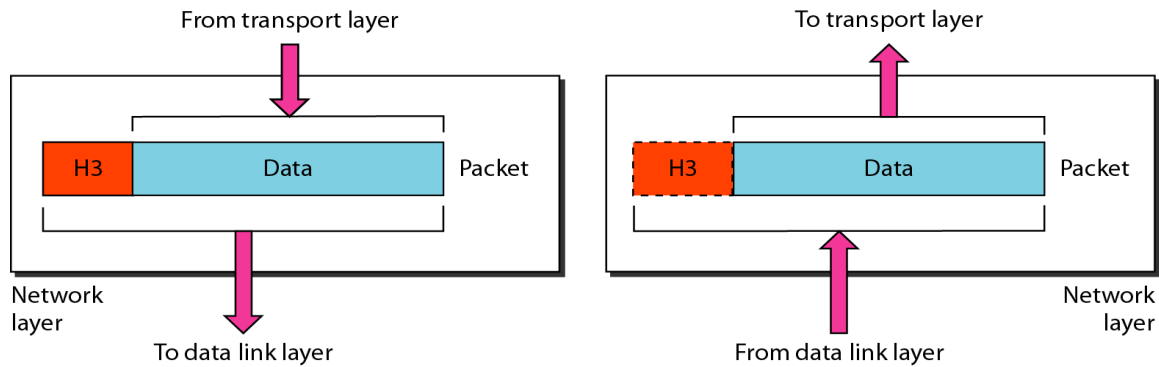
To make such end-to-end delivery possible, the network layer **provides** two related services: *Addressing and routing*.

Addressing end devices - In the same way that a phone has a unique telephone number, end devices must be configured with a unique IP address for identification on the network. An end device with a configured IP address is referred to as a host.

Routing means selecting the best path for sending a packet from one point to another, when more than one path is available.

Routing requires the addition of a header that includes, among other, information, the source and destination address of the packet. These addresses are different from the physical (node) addresses included in the data link header; it is known as a **logical address**.

Physical addresses are of current to next node only; they are changes as a frame move from one node to another. Whereas the logical addresses are those for the original source to the final destination, and do not change during packet transmission.

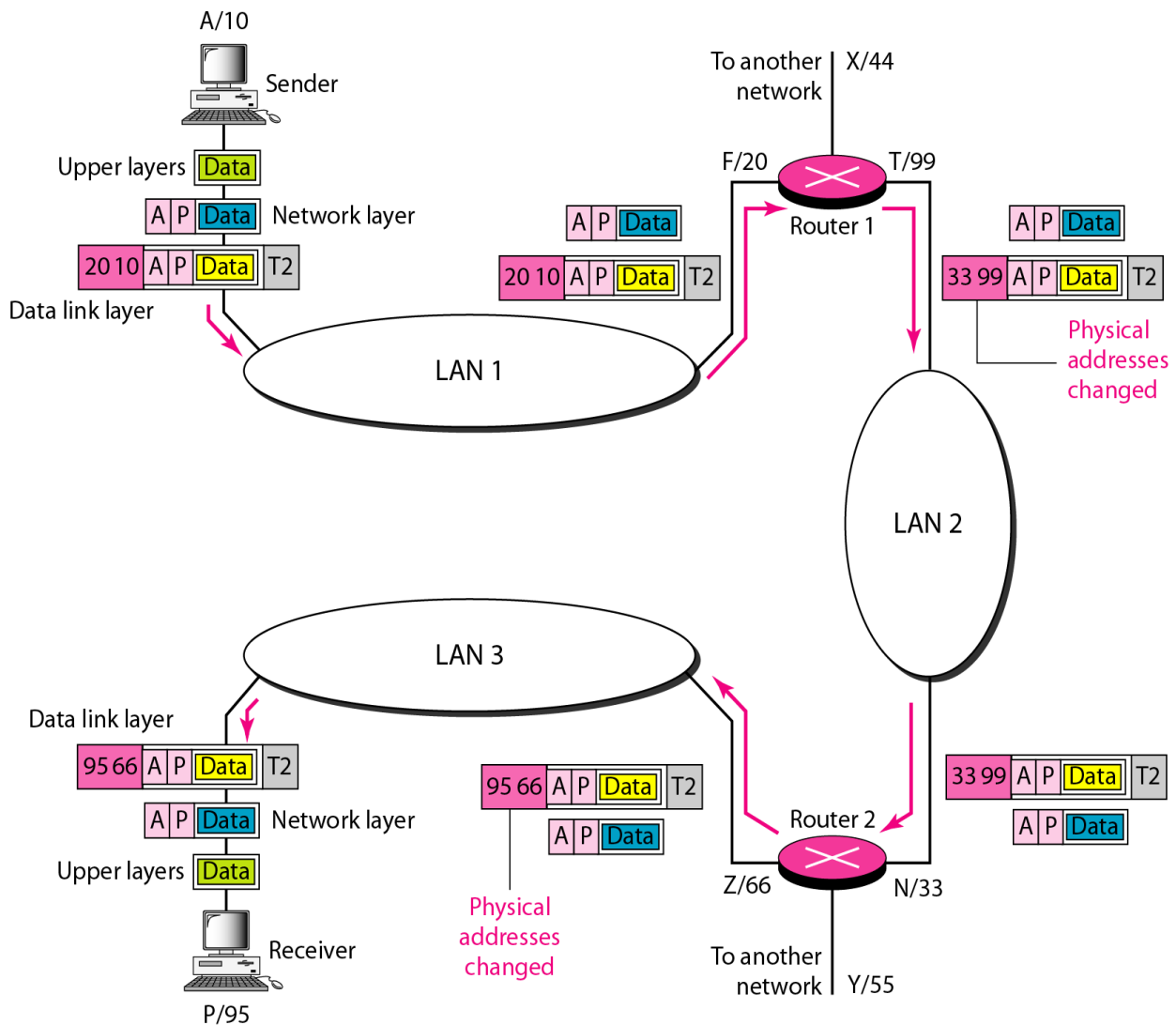


Specific responsibilities of the network layer include the following:

- ❖ **Source-to-destination delivery:** Moving a packet (best effort) from its point of origin to its intended destination across multiple network links.,
- ❖ **Logical addressing:** Inclusion of the source and destination addresses in the header.
- ❖ **Routing:** Deciding which of multiple paths a packet should take.
- ❖ **Address transformation:** Interpreting logical addresses to find their physical equivalents.

Example

Now imagine that in Figure below we want to send data from a node with network address **A**, physical address 10, located on one local area network, to a node with a network address **P**, physical address 95, located on another local area network. Because the two devices are located on different networks, we cannot use link addresses only. What we need here are universal addresses that can pass through the boundaries of local area networks. The network (logical) addresses have this characteristic. The packet at the network layer contains the logical addresses, which remain the same from the original source to the destination (**A** and **P** respectively, in the figure). They will not change when we go from network to network. However, the physical addresses will change when the packet moves from one network to another. The box with the **R** is a router (internetwork device), which we will discuss later



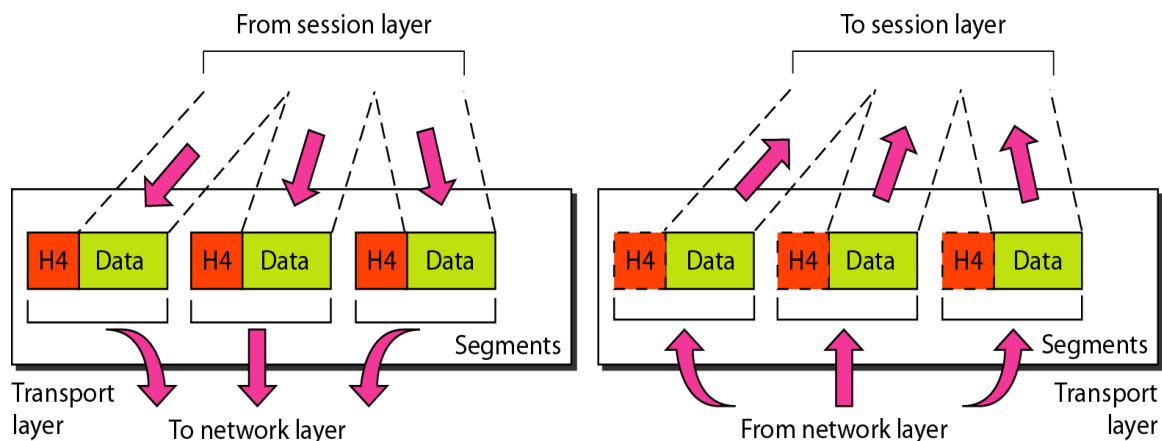
2.2.4- Transport Layer

The transport layer is **responsible for source -to-destination (end-to-end) delivery of the *entire message***. Whereas the network layer oversees end-to-end delivery of *individual packets*, it does not recognize any relationship between those packets. It treats each one independently as though each piece belonged to a separate message, whether or not it does. The transport layer, on the other hand, **ensures** that the whole message arrives intact and in order, overseeing both error control and flow control at the source-to-destination level.

Computers often run several programs at the same time. For this reason, source-to-destination delivery means delivery not only from one computer to the next but also from a specific application on one computer to a specific application on the other. The transport layer header must therefore include a type of address called a

service point address (also called a **port address** or socket address). The network layer gets each packet to the correct computer; the transport layer gets the entire message to the correct application on that computer.

As the transport layer receives the message from session layer it divides the message into a transmittable segment, indicating in the header the sequence of the segments so that it can be reassembled upon receipt at the destination.

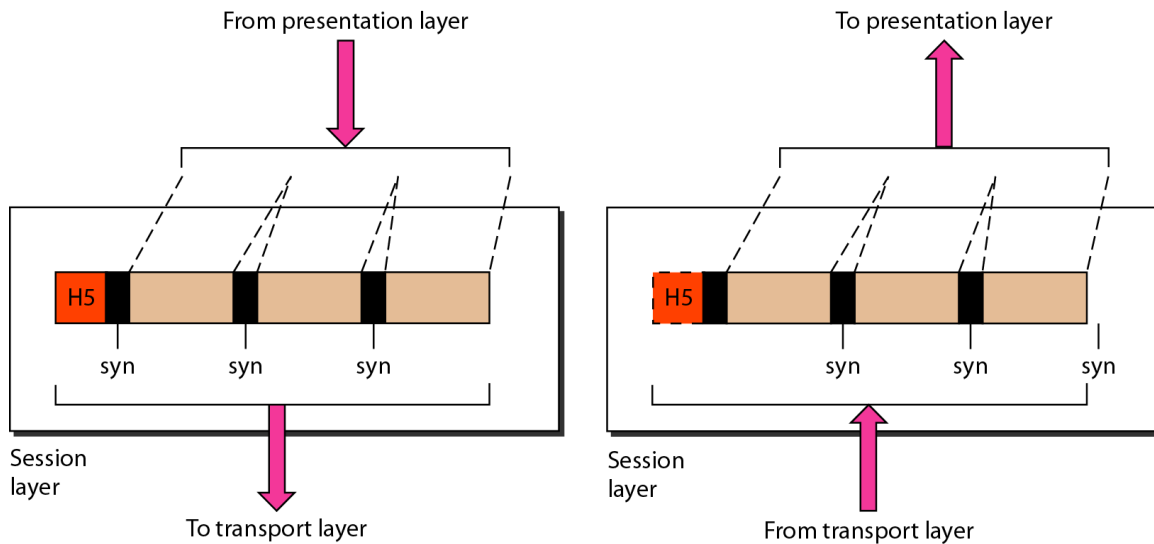


Specific responsibilities of the transport layer include the following:

- ❖ **End-to-end message delivery:** Overseeing the transmission and arrival of all packets of a message at the destination point.
- ❖ **Service-point (port) addressing:** Guaranteeing delivery of a message to the appropriate application on a computer running multiple applications.
- ❖ **Segmentation and reassembly:** Dividing a message into transmittable segments and marking each segment with a sequence number. These numbers enable the transport layer to reassemble the message correctly at the destination and to identify and replace packets lost in transmission.
- ❖ **Connection control:** Deciding whether or not to send all packets by a single path.

2.2.5- Session Layer

The session layer is the network *dialog controller*. It establishes, maintains, and synchronizes the interaction between communicating devices. It also ensures that each session closes appropriately rather than shutting down abruptly and leaving the user hanging.

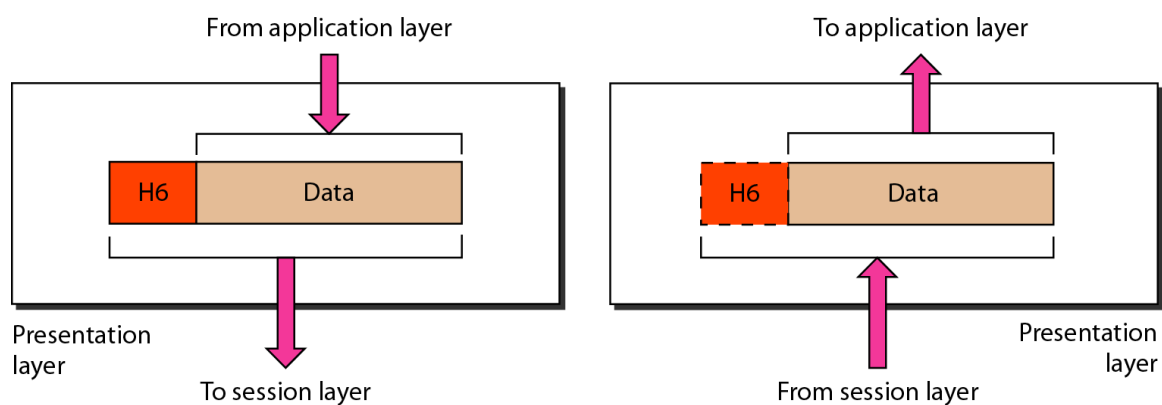


Specific responsibilities of the session layer include the following:

- ❖ **Session management:** Dividing a session into sub sessions by the introduction of checkpoints and separating long messages into shorter units called dialog units appropriate for transmission.
- ❖ **Synchronization:** Deciding in what order to pass the dialog units to the transport layer and where in the transmission to require confirmation from the receiver.
- ❖ **Dialog control:** Deciding who sends, and when.
- ❖ **Graceful close:** Ensuring that the exchange has been completed appropriately before the session closes.

2.2.6- Presentation Layer

The presentation layer **ensures** interoperability among communicating devices. Functions at this layer make it possible for two computers to communicate even if their internal representations of data differ (e.g., when one device uses one type of code and the other uses another). It **provides** necessary translation of different control codes and character sets, graphics characters, and so on to allow both devices to understand the same transmission the same way.

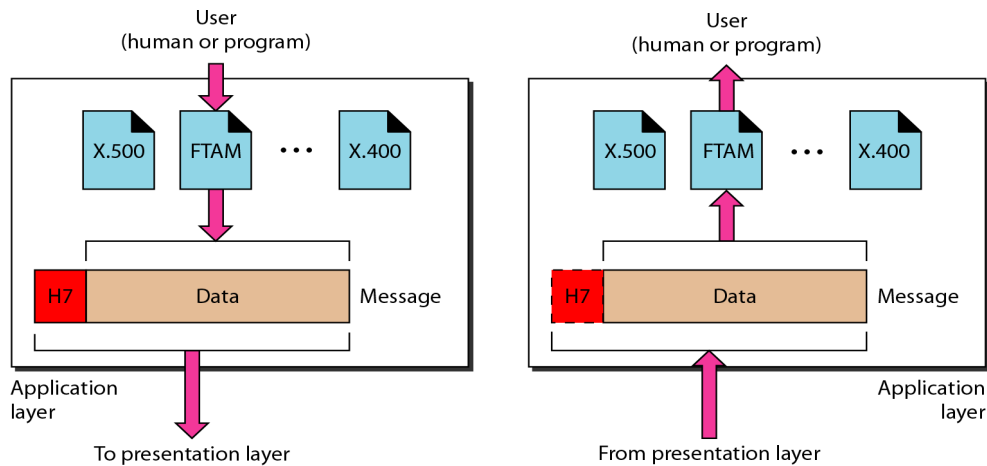


Specific responsibilities of the presentation layer include following:

- ❖ **Translation:** Changing the format of a message from that used by the sender into one mutually acceptable for transmission. Then, at the destination, changing that format into the one understood by the receiver.
- ❖ **Encryption:** Encryption and decryption of data for security purposes.
- ❖ **Compression:** Compressing and decompressing data to make transmission more efficient.
- ❖ **Security:** Validating passwords and log-in codes.

2.2.7- Application Layer

The application layer **enables** the user, whether human or software, to access the **network**. It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database management, and other types of distributed information services.



Specific services provided by the application layer include the following:

- ❖ **Web Services:** Allows accessing web content in the form of text, images and multimedia using the HTTP protocol
- ❖ **File access, transfer, and management:** Allows a user at a remote computer to access files in another host (to make changes or read data); to retrieve files from remote computer for use in the local computer; and to manage or control files in remote computer at that computer.
- ❖ **Mail services:** Provides the basis for electronic mail forwarding and storage.
- ❖ **Directory services:** Provides distributed database sources and access for global information about various objects.

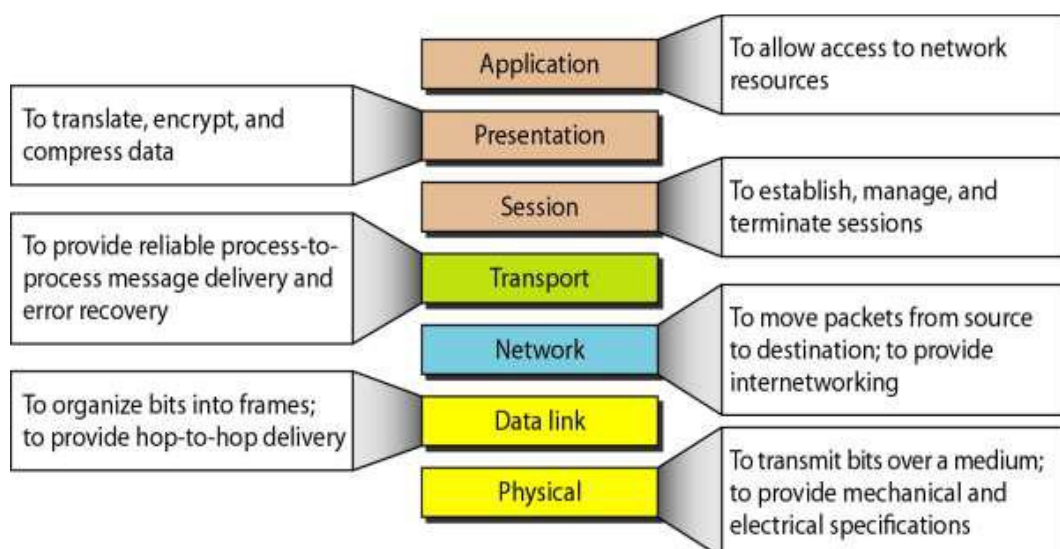
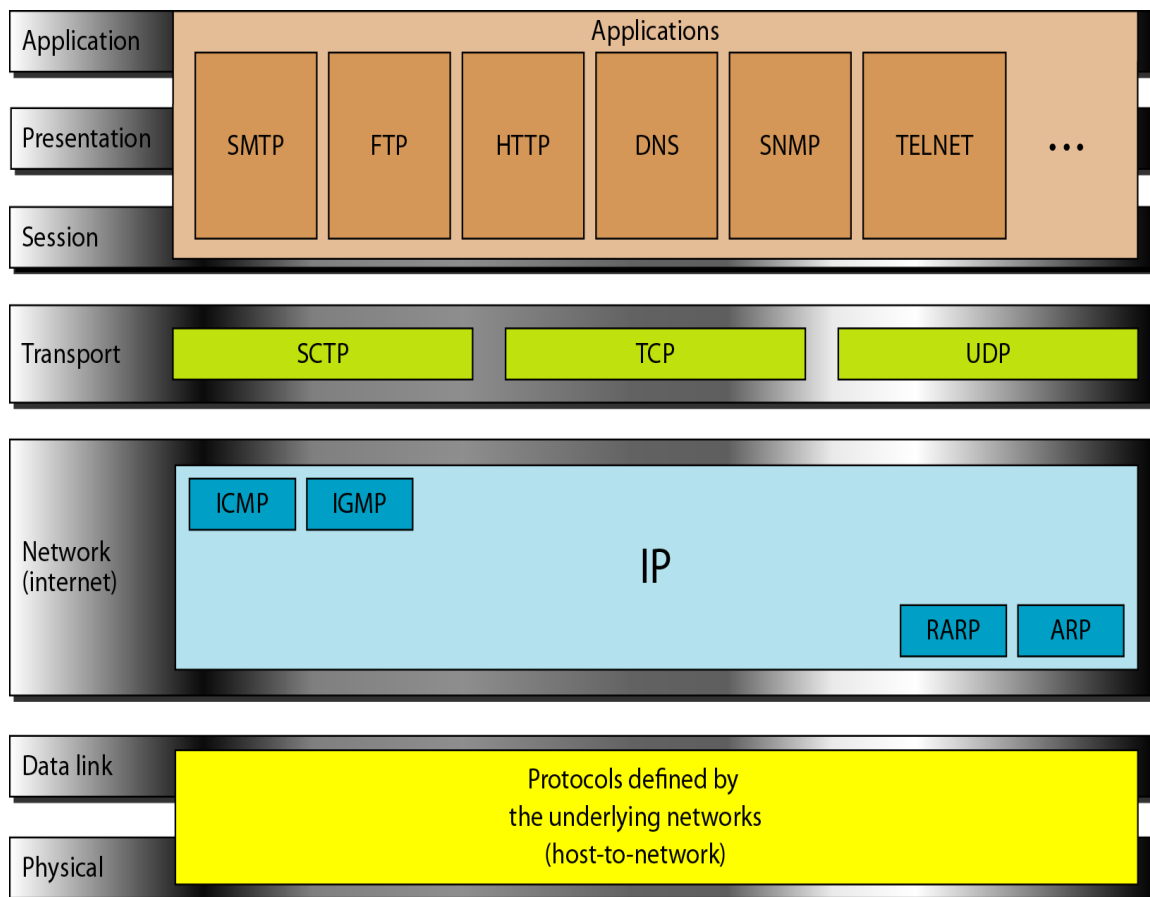


Figure (2.4) Summary of layers

2.3- TCP/IP Protocol Suite

The layers in the TCP/IP protocol suite do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having **four layers**: host-to-network, internet, transport, and application. However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: physical, data link, network, transport, and application, as shown in figure below.



The TCP/IP protocol stack is the de-facto standard in networking

- ✓ It is an alternative for the OSI 7-layer model, which has never really been implemented in practice.
- ✓ TCP/IP is an open standard and is the protocol used over the Internet.
- ✓ It can be found in most modern-day operating systems

The main differences between TCP/IP and the OSI 7-layer model are:

- Number of layers
 - TCP/IP defines only 4 or 5 layers.
- Functions performed at a given layer
 - In the OSI model each layer performs specific functions

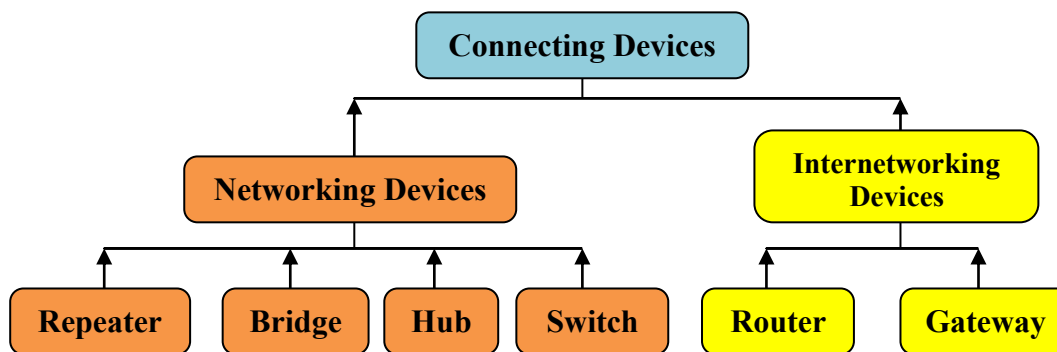
- In TCP/IP different protocols may be defined within a layer, each performing different functions. What is common about a set of protocols at the same layer is that they share the same set of support protocols at the next lower layer.

➤ Interface between adjacent layers

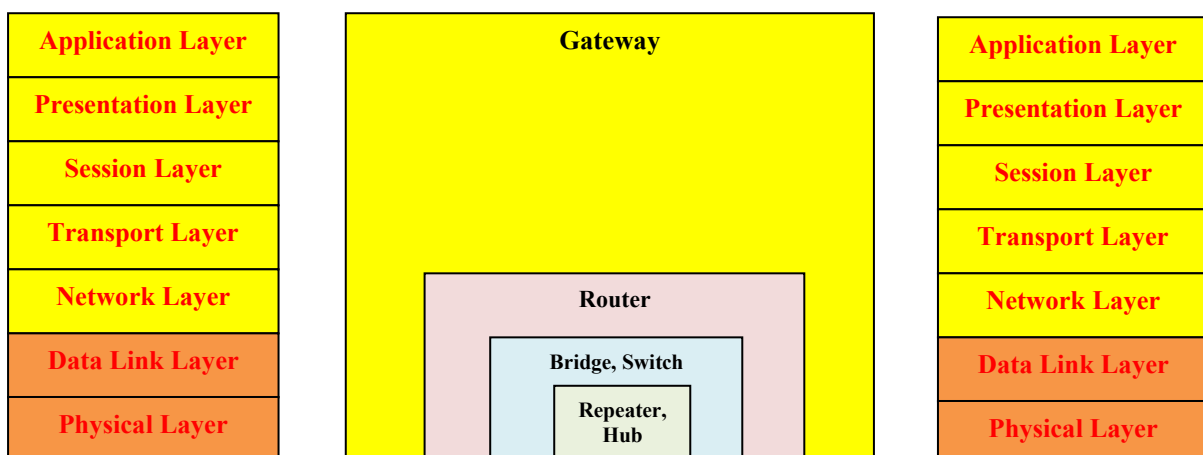
- In the OSI model, a protocol at a given layer may be substituted by a new one without impacting on adjacent layers.
- In TCP/IP the strict use of all layers is not mandated

2.4- Networking and Internetworks

To connect a network or multiple networks together there are different types of connecting devices as illustrated in figure below:

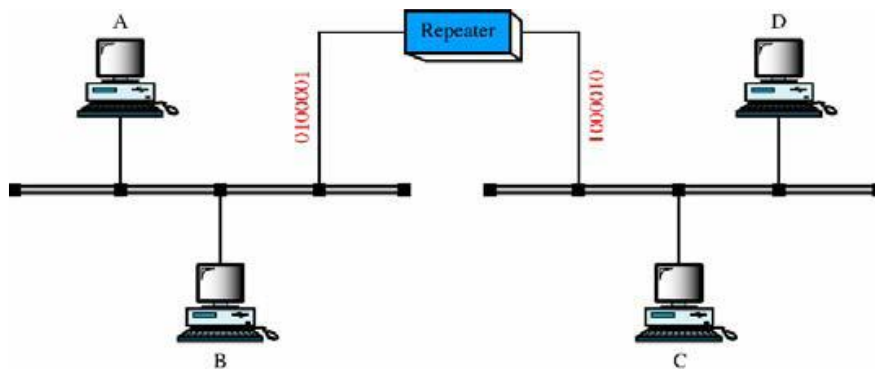


Each of these devices operates at different layer of the OSI model as shown in figure below:



2.4.1- Repeater:

Devices used to extend the network cable length beyond the limit of the specified cable. The purpose of a repeater is to **regenerate and retiming network signals** at the bit level to allow them to travel a longer distance on the media. Repeaters are classified as **Layer 1 devices** in the OSI model, because they act only on the bit level and look at no other information.

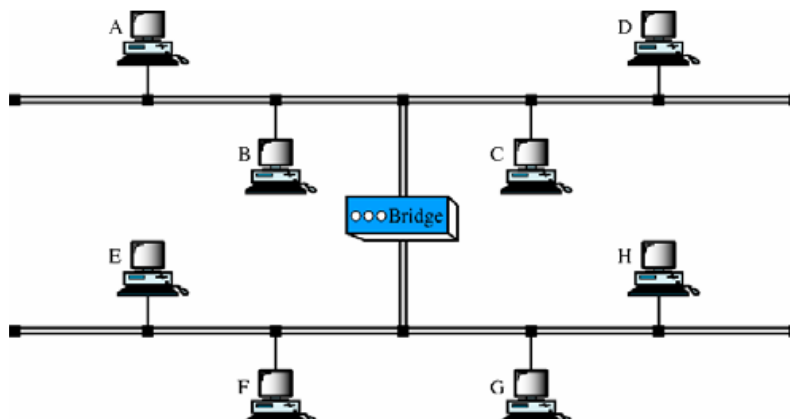


2.4.2- Hubs

The purpose of a hub is to **regenerate and retiming network signals**. This is done at the bit level to a large number of hosts (e.g. 4, 8, or even 24) using a process known as concentration. You will notice that this definition is very similar to the repeaters, which is why a hub is also known as *a multi-port repeater*. Hubs are considered **Layer 1 devices** because they only regenerate the signal and broadcast it out all of their ports (network connections).

2.4.3- Bridges

A bridge is a **Layer 2 device** designed to connect two LAN segments. The purpose of a bridge is to filter traffic on a LAN, to keep local traffic local, yet allow connectivity to other parts (segments) of the LAN for traffic that has been directed there.



2.4.4 Switches

A switch is a **Layer 2 device** just as a bridge is. In fact, a switch is called a ***multi-port bridge***, just like a hub is called a multi-port repeater. The difference between the hub and switch is that switches make decisions based on MAC addresses and hubs don't make decisions at all. Because of the decisions that switch make, they make a LAN much more efficient. They do this by "switching" data only out the port to which the proper host is connected. In contrast, a hub will send the data out all of its ports so that all of the hosts have to see and process (accept or reject) all of the data.

2.4.5- Routers

A router is a **Layer 3 device**. Working at Layer 3 allows the router to make decisions based on groups of network addresses as opposed to individual Layer 2 MAC addresses. Routers can also connect different Layer 2 technologies, such as Ethernet, Token-ring, and FDDI. However, because of their ability to route packets based on Layer 3 information, routers have become the backbone of the Internet, running the IP protocol. The purpose of a router is to examine incoming packets (Layer 3 data), choose the best path for them through the network, and then switch them to the proper outgoing port. A router is a type of internetworking device that passes data packets between networks, based on Layer 3 addresses. A router has the ability to make intelligent decisions regarding the best path for delivery of data on the network.



Republic of Iraq
Middle Technical University
Technical Engineering College of Artificial Intelligence
Cybersecurity Engineering Technology Department

Computer Network Fundamentals

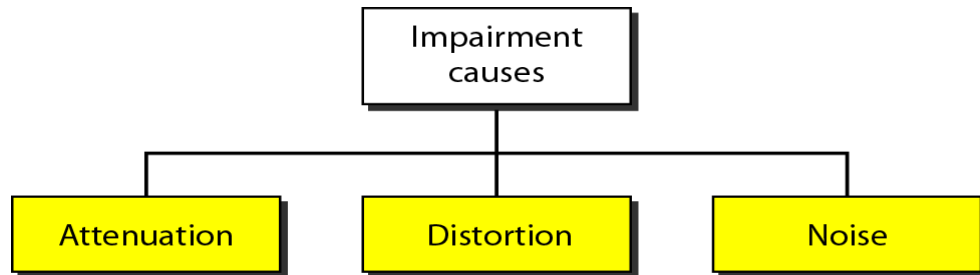
Chapter Three

Transmission Media & Multiplexing **((Physical Layer))**

Prepared By:
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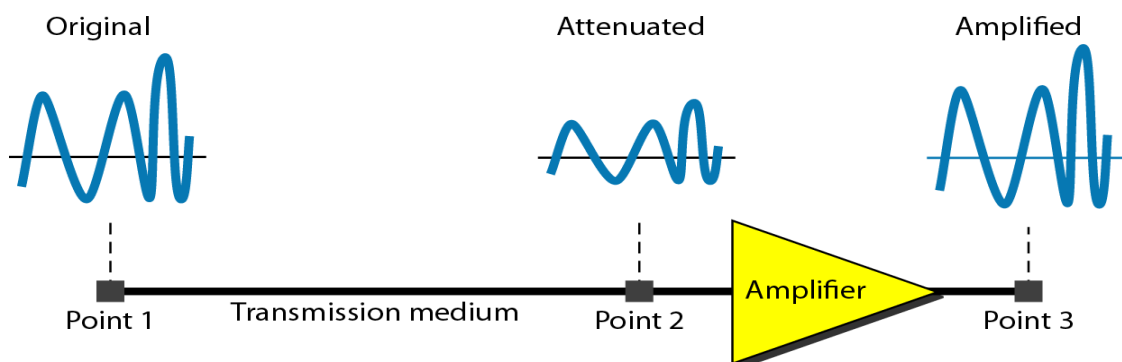
3.1- Transmission Impairment

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are *attenuation*, *distortion*, and *noise*



Attenuation

Attenuation means a loss of energy. When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium. That is why a wire carrying electric signals gets warm. Some of the electrical energy in the signal is converted to heat. To compensate for this loss, amplifiers are used to amplify the signal.



To show that a signal has lost or gained strength, engineers use the unit of the *decibel*. The **decibel (dB)** *measures the relative strengths of two signals or one signal at two different points.* Note that the decibel is negative if a signal is attenuated and positive if a signal is amplified.

$$\text{dB} = 10 \text{ Log}_{10} (P_2 / P_1)$$

Example:

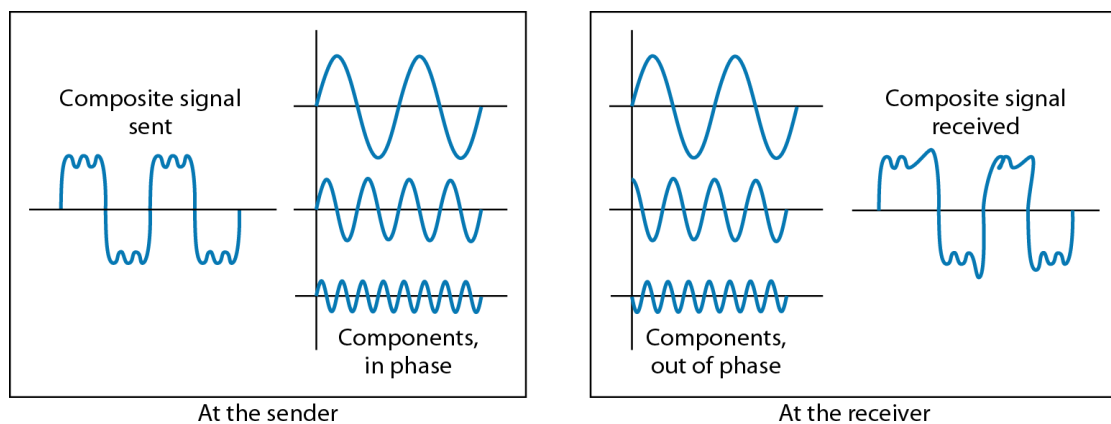
Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that P_2 is $(1/2)P_1$. In this case, the attenuation (loss of power) can be calculated as follows:

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (–3 dB) is equivalent to losing one-half the power.

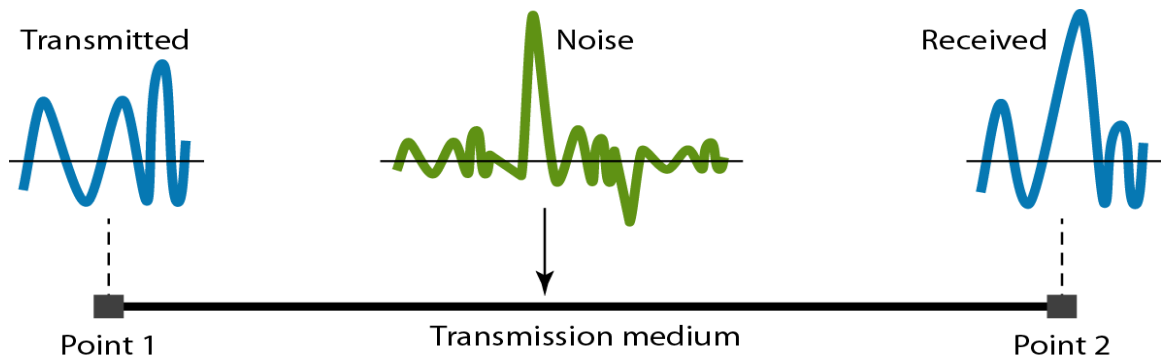
Distortion

Distortion means that the signal changes its form or shape. Distortion can occur in a composite signal made of different frequencies. Each signal component has its own propagation speed through a medium and, therefore, its own delay in arriving at the final destination. Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration. In other words, signal components at the receiver have phases different from what they had at the sender. The shape of the composite signal is therefore not the same.

**Noise**

Noise is another cause of impairment. Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal. Thermal noise is the random motion of electrons in a wire which creates an extra signal not originally sent by the transmitter. Induced noise comes from sources such as motors and appliances. These devices act as a sending antenna, and the transmission

medium acts as the receiving antenna. Crosstalk is the effect of one wire on the other. One wire act as a sending antenna and the other as the receiving antenna. Impulse noise is a spike (a signal with high energy in a very short time) that comes from power lines, lightning, and so on.



Signal-to-Noise Ratio (SNR)

It is the term useful to find the bit rate of the signal and it is defined as the ratio of the signal power to the noise power as shown below:

$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$

Because SNR is the ratio of two powers, it is often described in decibel units, SNR_{dB} and defined as:

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$$

Example:

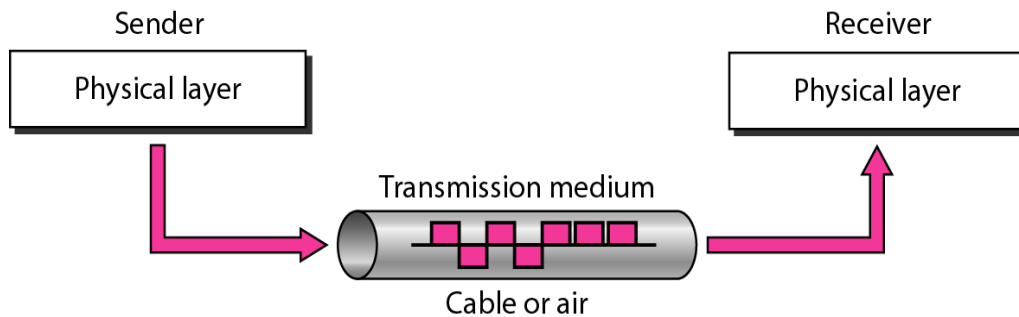
The power of a signal is 10 mW and the power of the noise is 1 μW ; what is the value of SNR_{dB}

$$\text{SNR} = (10000 \mu\text{W}) / (1 \mu\text{W}) = 10000$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} (10000) = 10 \log_{10} (10^4) = 40 \text{ dB}$$

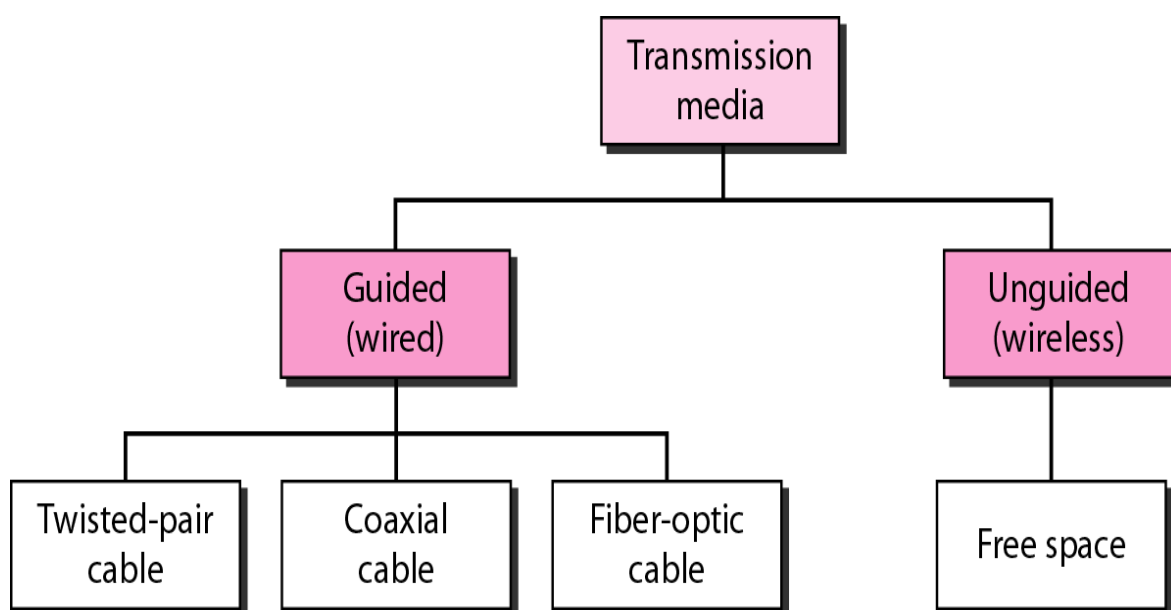
3.2- Transmission Media

We discussed many issues related to the physical layer in the previous sections. Now we will discuss transmission media. Transmission media are actually located below the physical layer and are directly controlled by the physical layer. You could say that transmission media belong to layer zero.



A transmission medium can be broadly defined as anything that can carry information from a source to a destination. In data communications the definition of the information and the transmission medium is more specific. The transmission medium is usually free space, metallic cable, or fiber-optic cable. The information is usually a signal that is the result of a conversion of data from another form. In telecommunications, transmission media can be divided into two broad categories: **guided** and **unguided**.

Guided media include twisted-pair cable, coaxial cable, and fiber-optic cable. Unguided medium is free space, as illustrated in the figure shown below.



3.2.1- Guided Media

Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable. A signal traveling along any of these media is directed and contained by the physical limits of the medium. Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current. Optical fiber is a cable that accepts and transports signals in the form of light.

Twisted Pair

A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together, as shown in Figure below.

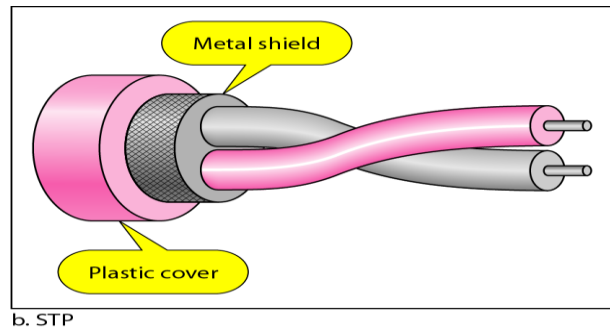
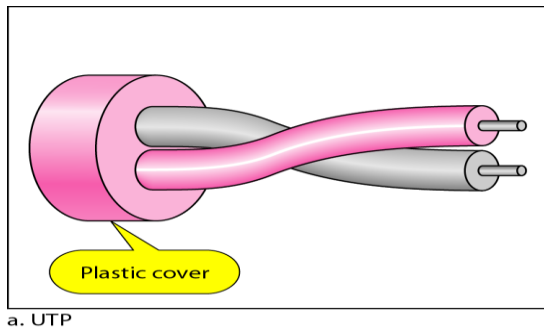


One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference. The receiver uses the difference between the two.

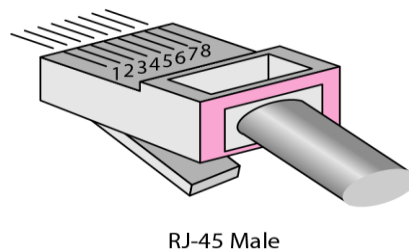
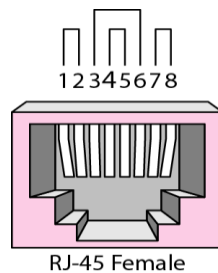
In addition to the signal sent by the sender on one of the wires, interference (noise) and crosstalk may affect both wires and create unwanted signals. Twisting makes it probable that both wires are equally affected by external influences (noise or crosstalk). This means that the receiver, which calculates the difference between the two, receives no unwanted signals. The unwanted signals are mostly canceled out.

Unshielded Versus Shielded Twisted-Pair Cable

The most common twisted-pair cable used in communications is referred to as unshielded twisted-pair (UTP). IBM has also produced a version of twisted-pair cable for its use called shielded twisted-pair (STP). STP cable has a metal foil or braided mesh covering that encases each pair of insulated conductors. Although metal casing improves the quality of cable by preventing the penetration of noise or crosstalk, it is bulkier and more expensive.

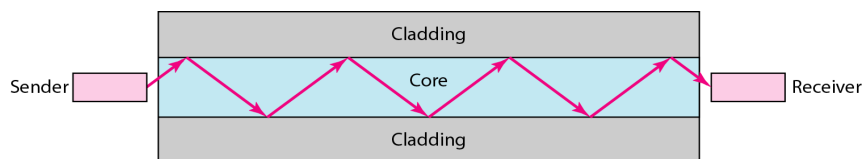


The most common UTP connector is RJ45 (RJ stands for registered jack), as shown below. The RJ45 is a keyed connector, meaning the connector can be inserted in only one way.

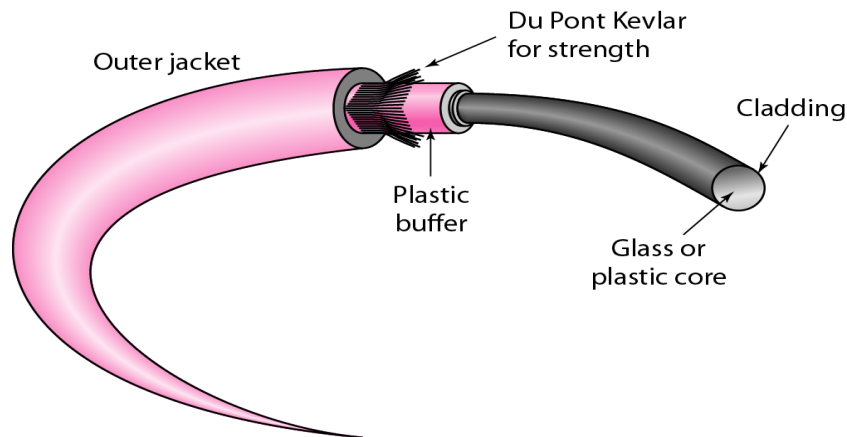


Fiber-Optic Cable

A fiber-optic cable is made of glass or plastic and transmits signals in the form of light. Optical fibers use reflection to guide light through a channel. A glass or plastic core is surrounded by a cladding of less dense glass or plastic. The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it, as shown in figure below.



Current technology supports two modes (multimode and single mode) for propagating light along optical channels, each requiring fiber with different physical characteristics.



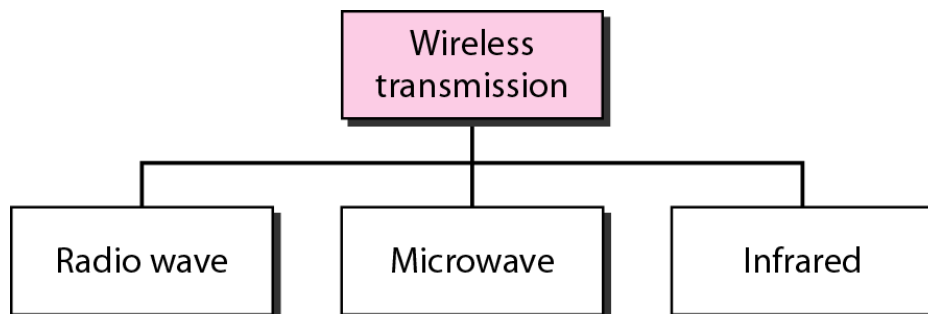
Different between single mode and multiple mode fiber optic

Single mode	Multiple mode
1:small core	1:larger core
2:less dispersion	2:greater dispersion and therefore ,loss of signal
3:suitable for long distant	3:large distant but shorter than single mode
4:use lasers as light source	4:use LED as source of light

3.2.2- Unguided Media: Wireless

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as *wireless communication*. Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

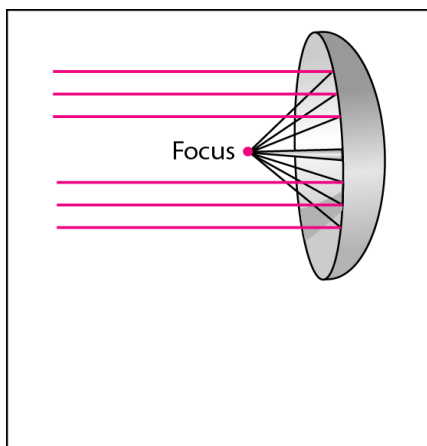
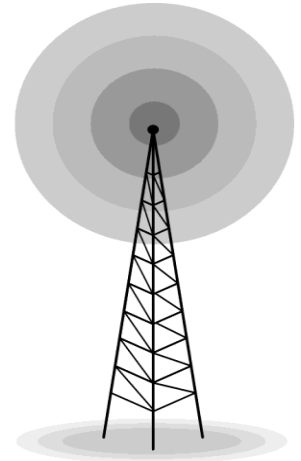
We can divide wireless transmission into three broad groups: **radio waves**, **microwaves**, and **infrared waves**.



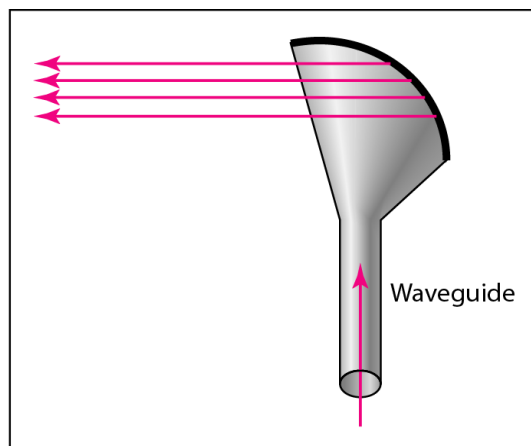
Radio Waves

Although there is no clear-cut demarcation between radio waves and microwaves, electromagnetic waves ranging in frequencies between **3 kHz and 1 GHz** are normally called radio waves; waves ranging in frequencies between 1 and 300 GHz are called microwaves.

However, the behavior of the waves, rather than the frequencies, is a better criterion for classification. Radio waves, for the most part, are ***Omni-directional***. When an antenna transmits radio waves, they are propagated in all directions. This means that the sending and receiving antennas do not have to be aligned. A sending antenna sends waves that can be received by any receiving antenna. While the Microwaves are ***unidirectional***. When an antenna transmits microwave waves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned.



a. Dish antenna



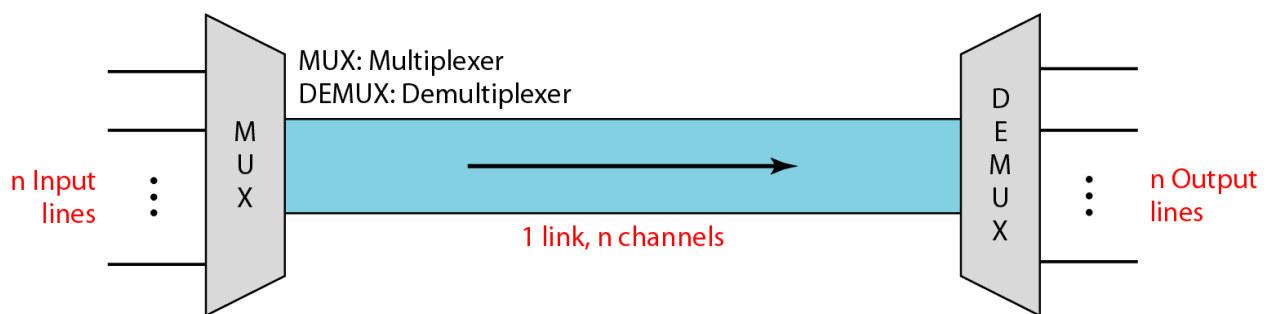
b. Horn antenna

3.3- Multiplexing

Whenever the transmission capacity of a medium linking two devices is greater than the transmission needs of the devices, the link **can be shared**, much as a large water pipe can carry water to several separate houses at once. *Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.*

As data and telecommunications usage increases, so does **traffic**. We can accommodate this increase by continuing to add individual lines each time a **new channel** is needed, **or** we can **install higher capacity links** and use each to carry multiple signals.

Today's technology includes **high-bandwidth media** such as *optical fiber*, and *terrestrial and satellite microwaves*. Each of these has a carrying capacity far in excess of that needed for the average transmission signal, **if the transmission capacity** of a link is greater than the transmission needs of the devices connected to it, the excess capacity is **wasted**. An efficient system maximizes the utilization of all facilities. In addition, the expensive technology involved often becomes cost-effective only when links are shared.



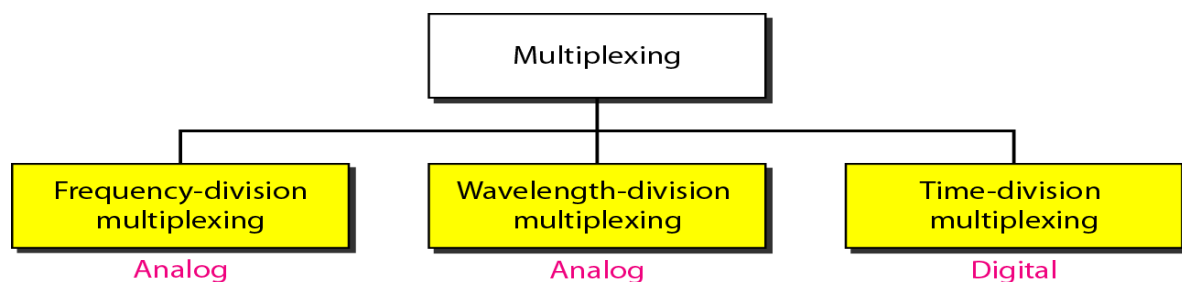
In a multiplexed system, ***n*** devices share the capacity of one link; The devices on the left direct their transmission streams to a **Multiplexer (MUX)** which combines them into a single stream (lots of to one). At the receiving end, that stream is fed into a **Demultiplexer (DEMUX)**, which separates the stream back into its component transmissions (one-to-many) and directs them to their intended receiving devices.

3.3.1- Types of Multiplexing

Signals are multiplexed using *three* basic techniques:

- **Frequency Division Multiplexing (FDM)**
- **Wavelength Division Multiplexing (WDM)**
- **Time Division Multiplexing (TDM)**

TDM is further subdivided into *synchronous TDM* (usually called TDM) and *asynchronous TDM*, also called statistical TDM or concentrator (See Figure (2))



Frequency-Division Multiplexing (FDM)

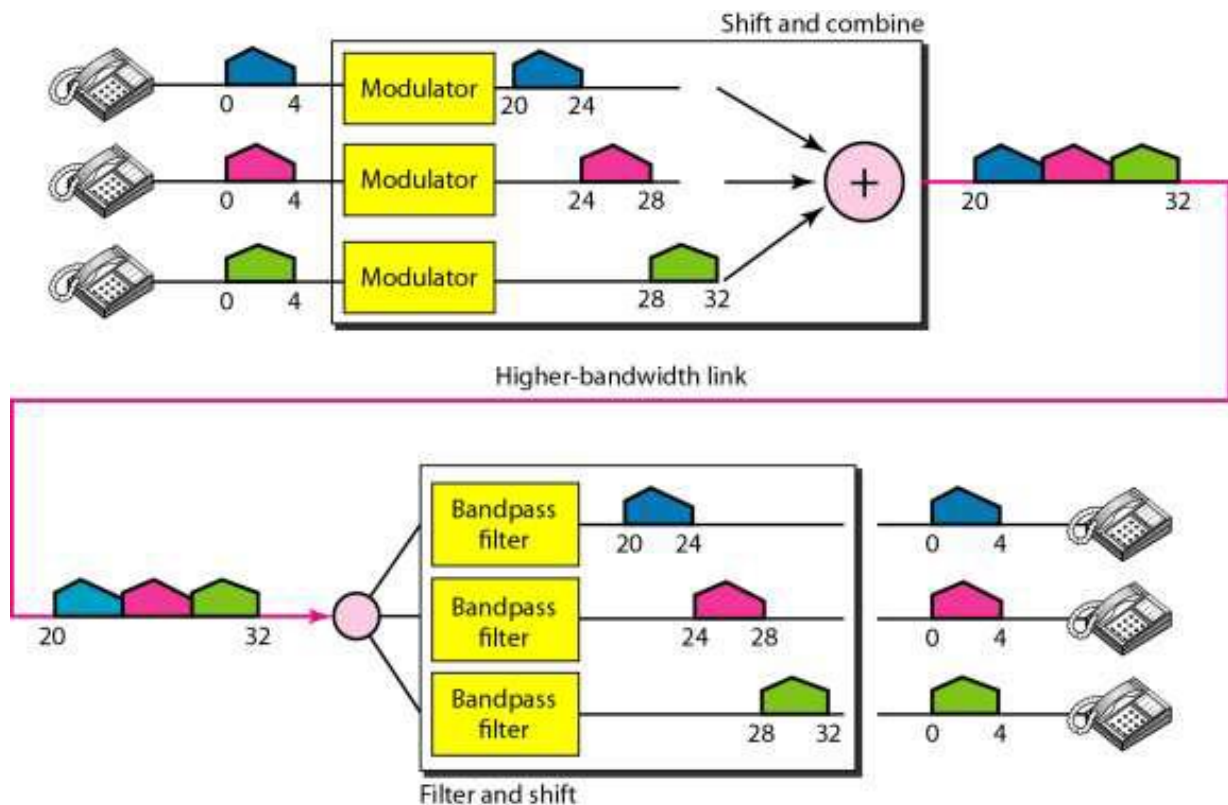
Frequency-division multiplexing (FDM) is an *analog technique* that can be applied when the bandwidth of a link is greater than the combined bandwidth of the signals to be transmitted. In FDM, *signals* generated by each sending device *modulate with* different carrier frequencies. These modulated signals are then *combined into* a single composite signal that can be transported by the link. Carrier frequencies are separated by enough bandwidth to accommodate the modulated signal. These bandwidth ranges are the channels through which the various signals travel. Channels must be separated by strips of unused bandwidth (*guard bands*) to prevent signals from overlapping. In addition, carrier frequencies must not interfere with the original data frequencies.



A familiar application of FDM is cable television. The coaxial cable used in a cable television system. Today, a new and more efficient method is being developed to implement FDM over fiber-optic cable. Called **Wavelength Division Multiplexing (WDM)**, it uses essentially the same concepts as FDM but incorporates the range of frequencies in the visible light spectrum.

Example 1

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the multiplexer process and de multiplexer process in frequency domain (configuration using the frequency domain). Assume there are no guard bands.



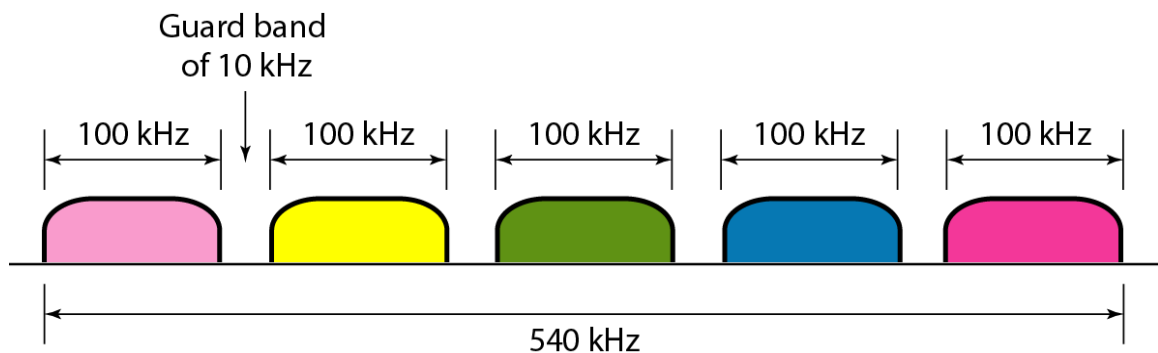
Example 2

Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

Solution

For five channels, we need at least four guard bands. This means that the required bandwidth is at least as shown in Figure below.

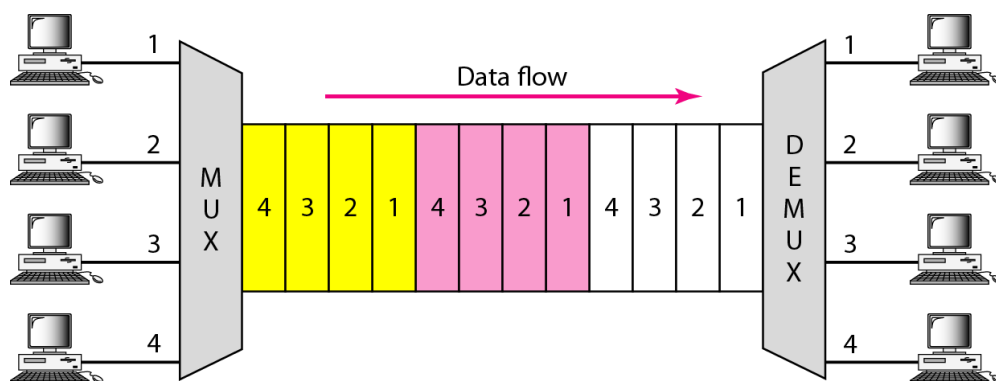
$$5 \times 100 + 4 \times 10 = 540 \text{ kHz},$$



Time-Division Multiplexing (TDM)

Time-division multiplexing (TDM) is a **digital process** that can be applied when the data rate capacity of the transmission medium is greater than the data rate required by the sending and receiving devices. In such a case, multiple transmissions can occupy a single link by *subdividing* them and *interleaving* the portions.

Note that the same link is used as in the FDM. Here however, the link is shown *sectioned by time rather than frequency*.



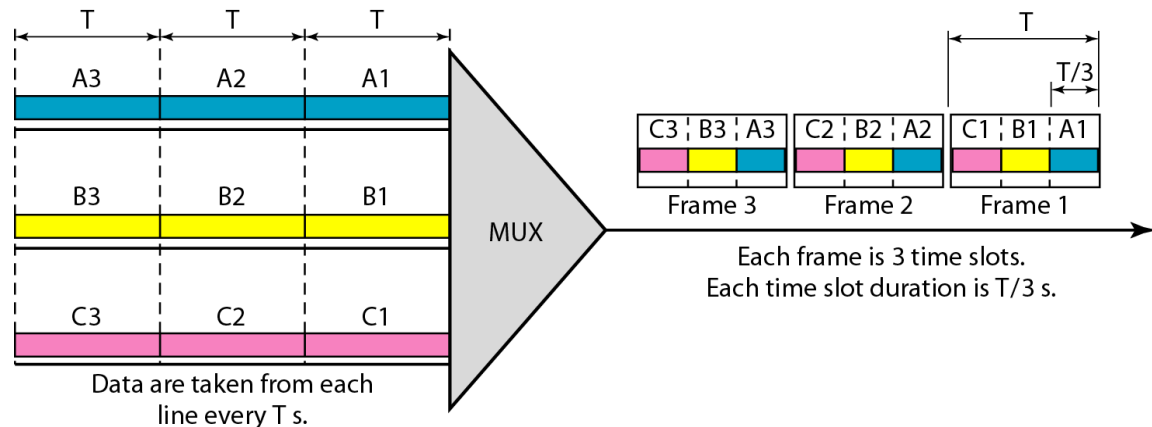
TDM can be implemented in two ways: **synchronous TDM** and **asynchronous TDM**.

a) Synchronous TDM

Here *synchronous means that the multiplexer allocates exactly the same time slot to each device at all times, whether or not a device has anything to transmit*. **Time slot A**, for example, is assigned to device **A** alone and cannot be used by any other device. Each time its allocated time slot comes up, a device has the opportunity to send a portion of its data. If a device is unable to transmit or does not have data to send, its time slot remains **empty**.

Time slots are grouped into frames. In a system with n input lines, each frame has at least n slots. With each slot allocated to carry data from a specific input line. If all the input devices sharing a link are transmitting at the *same data rate*, each device has *one time slot per frame*. However, *it is possible to* accommodate varying data rates. A transmission with two slots per frame will arrive twice as quickly as one with one slot per frame.

In Figure below, we show three input lines multiplexed onto a single path using synchronous TDM. In this example, all of the inputs have the same data rate, so the number of time slots in each frame is equal to the number of input lines.



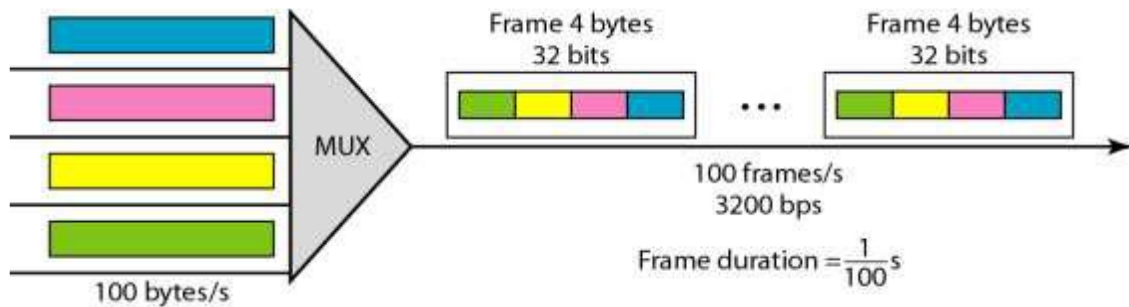
Example 3

Four channels are multiplexed using TDM. If each channel sends 100 bytes/sec and we multiplex 1 byte per channel.

- 1- Show the frame traveling on the link
- 2- The size of the frame
- 3- The frame rate
- 4- The duration of a frame
- 5- Bit rate for the link.

Solution

1-



2- Size of frame=input multiplexer*number of channels

$$\text{Frame size} = 1 * 4 = 4 \text{ byte or } 32 \text{ bits}$$

3-Because each channel is sending 100 bytes/s and a frame carries 1 byte from each channel, the frame rate must be 100 frames per second.

4-The duration of a frame is therefore $\frac{1}{100} \text{ s}$.

5-The link is carrying 100 frames per second, and since each frame contains 32 bits, the bit rate is 100×32 , or 3200 bps.

This is actually 4 times the bit rate of each channel, which is $100 \times 8 = 800 \text{ bps}$.

Example 4

Four channels are multiplexed using TDM. If each channel sends 1k bits/s and we multiplex 1 bit per channel find:

- (a) The input bit duration.
- (b) The output bit rate.
- (c) The output bit duration.
- (d) The output frame rate.

Solution

We can answer the questions as follows:

- a. The duration of 1 bit before multiplexing is $\frac{1}{1 \text{ kbps}}$, or 0.001 s (1 ms).
 - b. The rate of the link is 4 times the rate of a connection, or 4 kbps.
 - c. The duration of each time slot is one-fourth of the duration of each bit before multiplexing, or $\frac{1}{4} \text{ ms}$ or 250 μs .
- Note that we can also calculate this from the data rate of the link, 4 kbps. The bit duration is the inverse of the data rate, or $\frac{1}{4 \text{ kbps}}$ or 250 μs .
- d. The frame rate is always the same as any input rate = 1k bps.

Example 5

Four channels synchronous TDM with a data stream for each input 1 Mbps and the unit of data is 1 bit. Find:

- (a) The input bit duration
- (b) The output bit duration

- (c) The output bit rate
- (d) The output frame rate.

Solution

- a. The input bit duration = $1/\text{input bit rate} = 1/1 \text{ Mps} = 1 \mu\text{s}$.
- b. The output bit duration = $1/\text{number of input channel} = 1/4 \mu\text{s}$.
- c. The output bit rate = $\text{number of channel} * \text{rate of channel} = 1 \text{ Mbps} * 4 = 4 \text{ Mbps}$ or it is the inverse of the output bit duration or $1/(1/4 \mu\text{s})$ or 4 Mbps.
- d. The frame rate is always the same as any input rate = 1 Mbps.

b) Asynchronous TDM

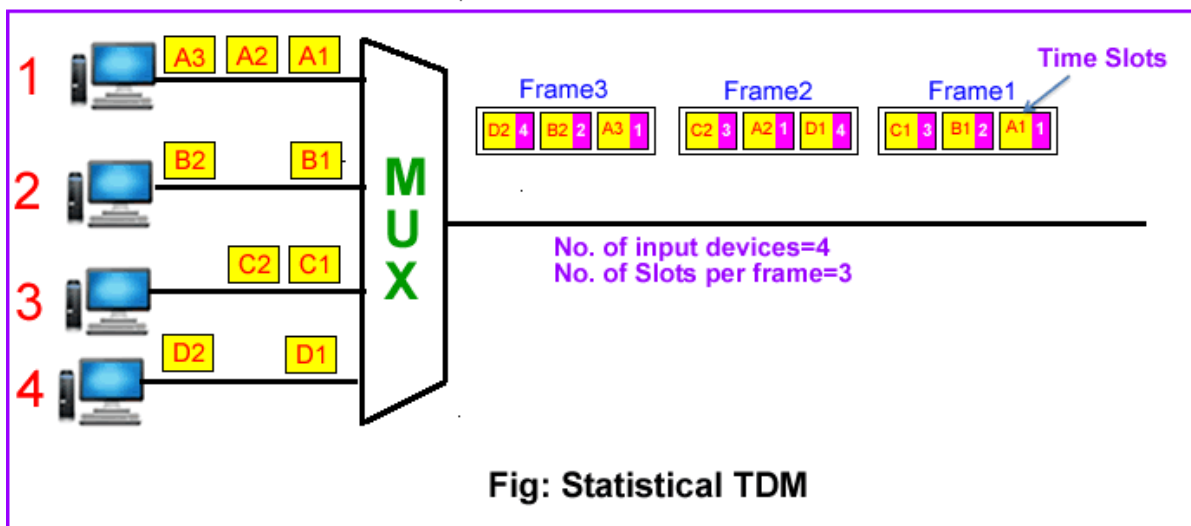
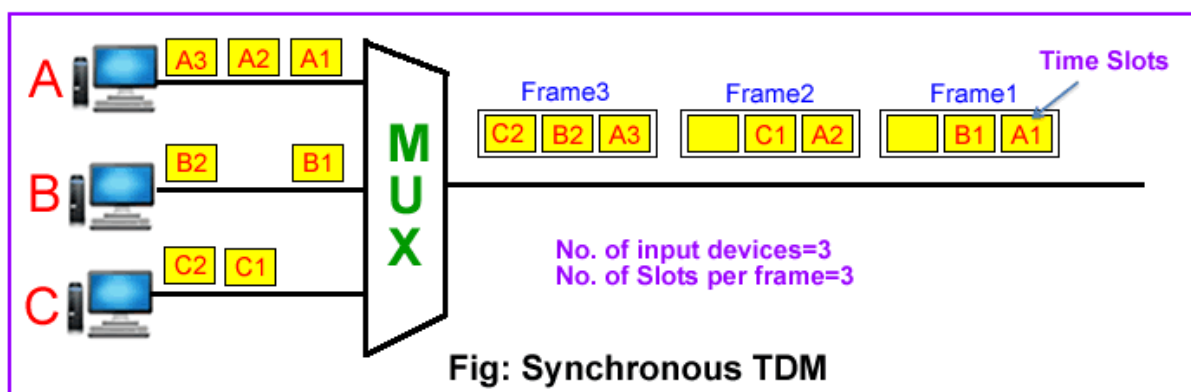
Synchronous TDM *does not guarantee* that the full capacity of a link is used. In fact, it is more likely that only a portion of the time slots is in use at a given instant. **Because the time** slots are pre-assigned and fixed whenever a connected device is not transmitting the corresponding slot is **empty** and that much of the path is **wasted**. For example, imagine that we have multiplexed the output of 20 identical computers onto a single line. Using synchronous TDM, the speed of that line must be at least 20 times the speed of each input line. But what if only 10 computers are in use at a time? Half of the capacity of the line is wasted.

Asynchronous time-division multiplexing, or statistical time-division multiplexing, is designed to avoid this type of waste. As with the term *synchronous*, the term **asynchronous means** something different in multiplexing than it means in other areas of data communications. Here it means flexible or not fixed.

Like synchronous TDM, asynchronous TDM allows a number of lower speed input lines to be multiplexed to a single higher speed line. **Unlike** synchronous TDM, however, in asynchronous TDM the total speed of the input lines can be greater than the capacity of the path. In a **synchronous system**, if we have ***n*** input lines, the frame contains a fixed number of at least ***n*** time slots. In an asynchronous system, if we have ***n*** input lines, the frame contains no more than ***m*** slots, with ***m*** less than ***n***. In this way, asynchronous TDM supports the same number of input lines as synchronous TDM with a lower capacity link. Or given the same link, asynchronous TDM can support more devices than synchronous TDM.

The number of time slots in an asynchronous TDM frame (m) is based on a statistical analysis of the number of input lines that are likely to be transmitting at any given time. Rather than being pre-assigned, each slot is available to any of the attached input that has data to send. The multiplexer scans the input line, accept portions of the data until a frame is filled, and then sends the frame across the link. If there are not enough data to full all the slots in a frame, the frame is transmitted only partially filled; thus, full link capacity may not be used 100 percent of the time.

Figure below illustrate the difference in operation between synchronous and asynchronous TDM.





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Computer Network Fundamentals

Chapter Four

Physical Addressing & Error Detection **((Data Link Layer))**

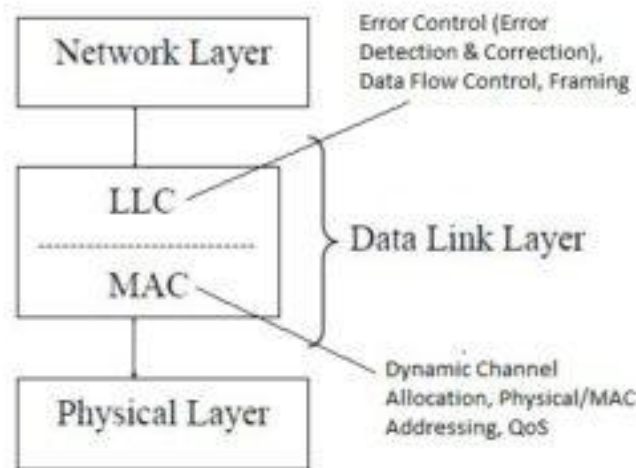
Prepared By:
Associate Prof. Dr. Mahmoud Shuker Mahmoud
mahmoud.shukur@mtu.edu.iq

4.1- Introduction

The data link layer is structured into two sublayers: the **Media Access Control (MAC) sublayer** and the **Logical Link Control (LLC) sublayer**.

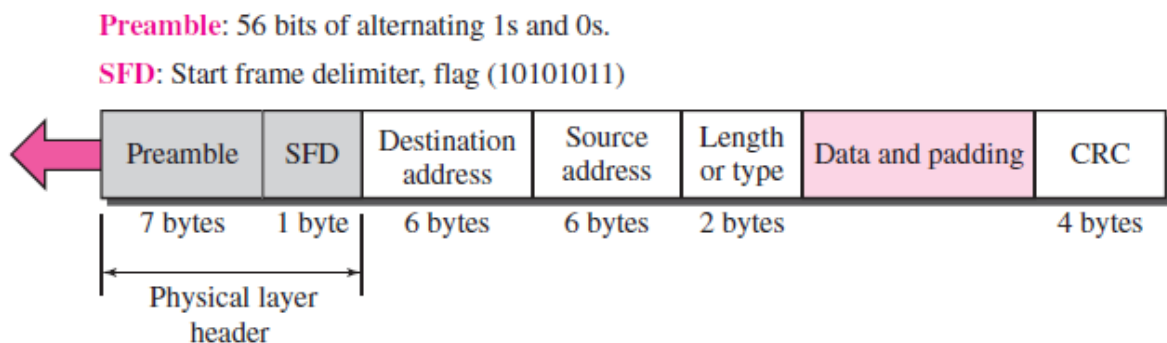
The MAC sublayer controls access to the physical network medium using MAC addresses, handling data encapsulation and managing access to avoid collisions.

The LLC sublayer provides flow control, error detection, and synchronization, and determines the protocol being used.



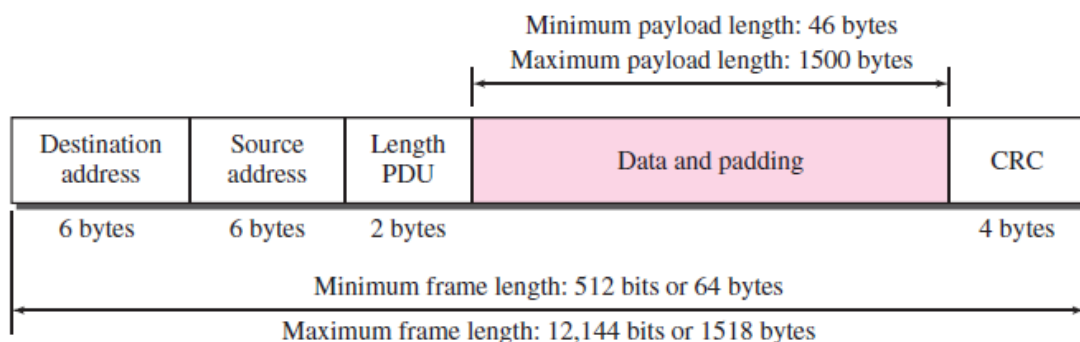
4.2- Frame Format

The Ethernet frame contains seven fields: preamble, SFD, DA, SA, length or type of protocol data unit (PDU), upper-layer data, and the CRC. The format of the MAC frame is shown in Figure below.



- **Preamble:** The first field of the frame contains 7 bytes (56 bits) of alternating 0s and 1s that alerts the receiving system to the coming frame and enables it to synchronize its input timing. The preamble is actually added at the physical layer and is not (formally) part of the frame.
- **Start frame delimiter (SFD):** The second field (1 byte: 10101011) signals the beginning of the frame. The SFD warns the station or stations that this is the last chance for synchronization. The last 2 bits is 11 and alerts the receiver that the next field is the destination address.
- **Destination address (DA):** The DA field is 6 bytes and contains the physical address of the destination station or stations to receive the packet.
- **Source address (SA):** The SA field is also 6 bytes and contains the physical address of the sender of the packet.
- **Length or type:** This field is defined as a type field or length field. The original Ethernet used this field as the type field to define the upper-layer protocol using the MAC frame. The IEEE standard used it as the length field to define the number of bytes in the data field. Both uses are common today.
- **Data:** This field carries data encapsulated from the upper-layer protocols. It is a minimum of 46 and a maximum of 1500 bytes.
- **CRC:** The last field contains error detection information.

Ethernet has imposed restrictions on both the minimum and maximum lengths of a frame, as shown in Figure below.



4.3- Physical Addressing

Each station on an Ethernet network (such as a PC, workstation, or printer) has its own network interface card (NIC). The NIC fits inside the station and provides the station with a 6-byte physical address (MAC), normally written in hexadecimal notation.

06 : 01 : 02 : 01 : 2C : 4B

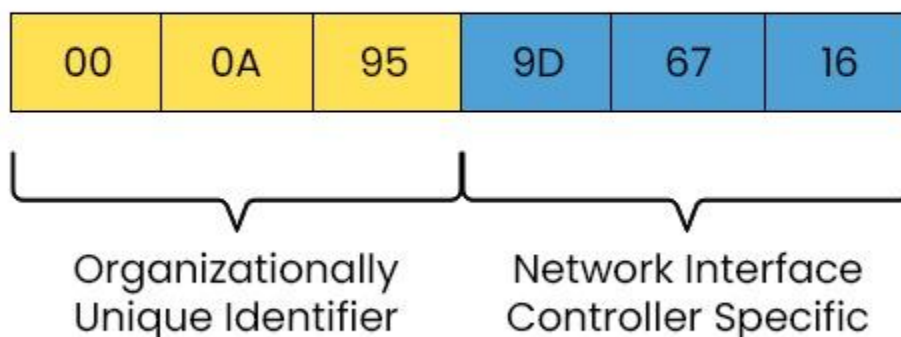
6 bytes = 12 hex digits = 48 bits

A MAC address is a 48-bit hardware identifier, represented as 12 hexadecimal digits, that is divided into two 24-bit halves.

- The first half, the **Organizationally Unique Identifier (OUI)**, identifies the manufacturer.
- The second half is a **unique device identifier** assigned by the manufacturer to the NIC.

These are typically displayed as six groups of two digits, separated by colons or hyphens (e.g., 00:0A:95:9D:67:16 **or** 00-0A-95-9D-67-16).

Media Access Control Address



4.4- Error Handling

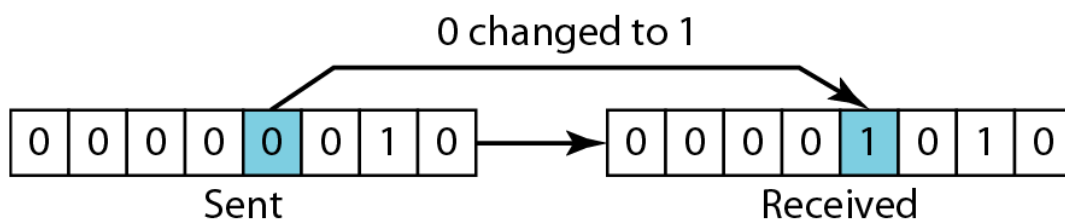
Networks must be able to transfer data from one device to another with *complete accuracy*. Yet any time data are transmitted from source to destination, they can become corrupted in passage. In fact, it is more likely that some part of a message will be altered in transit than that the entire contents will arrive intact. Reliable systems must have a mechanism for detection, and correcting such errors.

4.4.1- Types of Errors

Whenever an electromagnetic signal flows from one point to another, it is subject to *unpredictable interference from heat, magnetism, and other forms of electricity*. This interference can change the **shape** or **timing** of the signal. Such changes can alter the meaning of the data. Changing 0 to 1 or 1 to 0. Bits can be changed singly or in clumps. So it is important to understand three types of errors. These types are usually referred to as **single-bit**, **multiple-bit**, and **burst errors**. Of the three, a single-bit, error is the most likely to occur and a burst error the least likely.

a) Single Bit Error

The term single-bit error means that only one bit of a given data unit (such as a byte, character, data unit or packet) is changed from 1 to 0 or from 0 to 1.

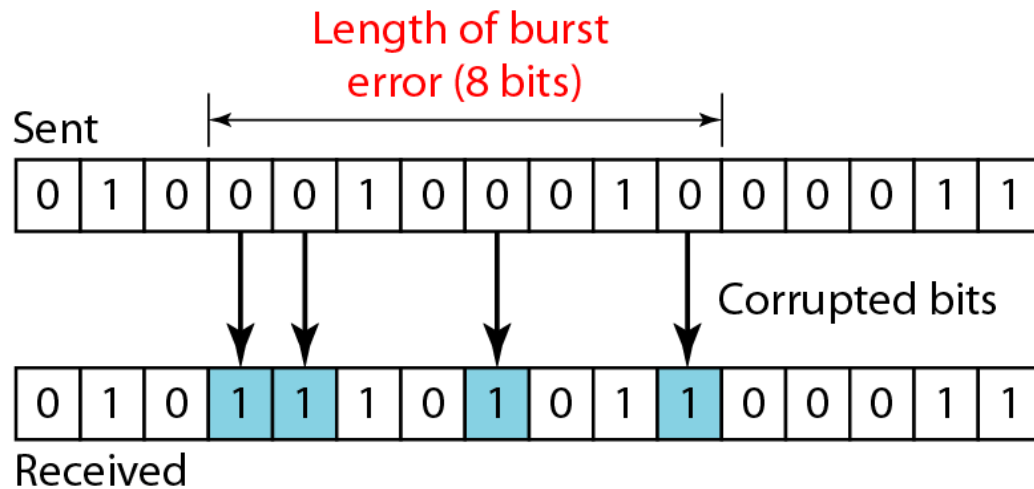


b) Multiple-Bit Error

The term multiple-bit error means that two or more non-consecutive bits in a data unit have changed from 1 to 0 or from 0 to 1.

c) Burst Error

The term burst error means that two or more consecutive bits in the data unit have changed from 1 to 0 or from 0 to 1. In this case, 0100010001000011 was sent, but 0101101101000011, was received.

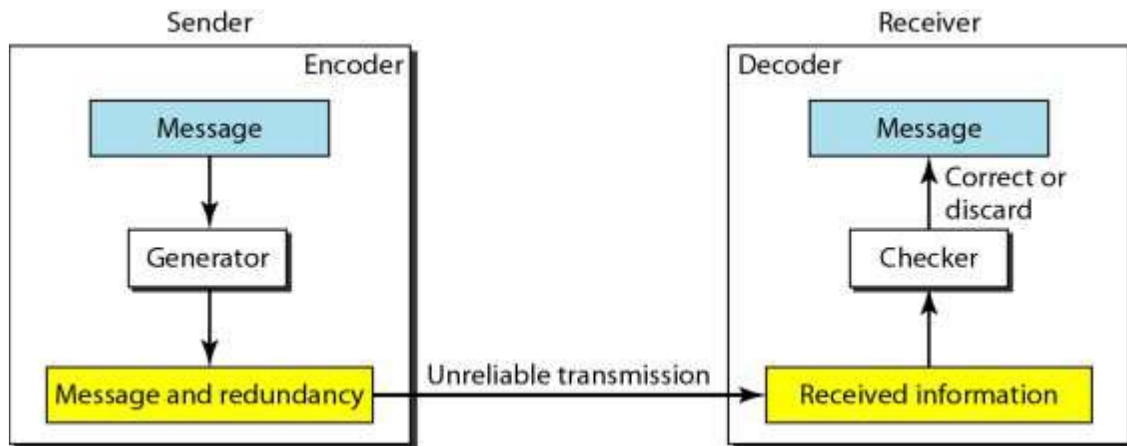


4.4.2- Error Detection

For a machine to check for errors this way would be slow, costly, and of questionable value. What we need is a mechanism that is simple and completely objective.

One mechanism that would satisfy these requirements would be to *send every data unit twice*. The receiving device would then be able to do a bit for bit comparison between the two versions of the data. Any discrepancy would indicate an error and an appropriate correction mechanism could be set in place. This system would be completely accurate (the odds of errors being introduced onto exactly the same bits in both sets of data are infinitesimally small), but it would also be insupportably slow. Not only would the transmission time double, but the time it takes, to compare every unit bit by bit must be added.

The concept of including extra information solely for the purposes of comparison is a good one. But instead of repeating the entire data stream, a shorter group of bits may be appended to the end of each unit. This technique is called **redundancy** because the extra bits are redundant to the information; they are discarded as soon as the accuracy of the transmission has been determined.



Four types of redundancy checks are used in data communications:

- a. Vertical Redundancy Check (VRC) (also called parity check).
- b. Longitudinal Redundancy Check (LRC).
- c. Cyclical Redundancy Check (CRC), and
- d. Checksum.

Cyclical Redundancy Check (CRC)

The most powerful redundancy technique, CRC is based on binary division. The redundancy bits used by CRC are derived by dividing the data unit by the predetermined divisor and appending it to the end of the data string so that the resulting bit sequence must be exactly divisible by the divisor.

The CRC generator uses modulo-2 division. In the first stage, the four-bit divisor is subtracted from the first four bits of the dividend. Each bit of the divisor is subtracted from the corresponding bit of the dividend without disturbing the next higher bit.

The CRC generator is most often represented as an algebraic polynomial. This is useful because the code is short and can be easily mathematically understood.

$$x^7 + x^5 + x^2 + x + 1$$

$$\Downarrow$$

$$10100111$$

Gives the binary:

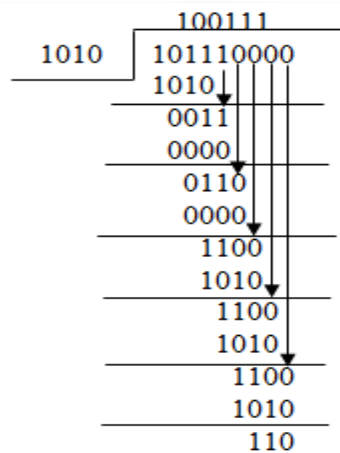
Example: Determine the CRC code to be used to send message $M(x)$ by using the divisor $G(x)$ given:

$$M(x) = X^5 + X^3 + X^2 + X$$

$$G(x) = X^3 + X$$

Form the polynomial $M(x) = 101110$
And the $G(x) = 1010$

Sender Side

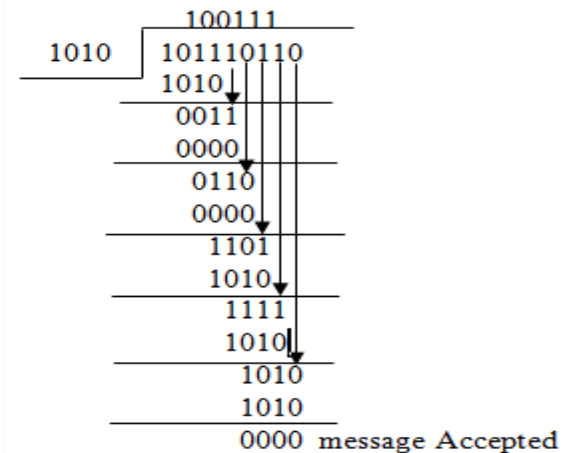


The CRC code is=110

And

The message to be send= 101110110

Receiver Side



Checksum

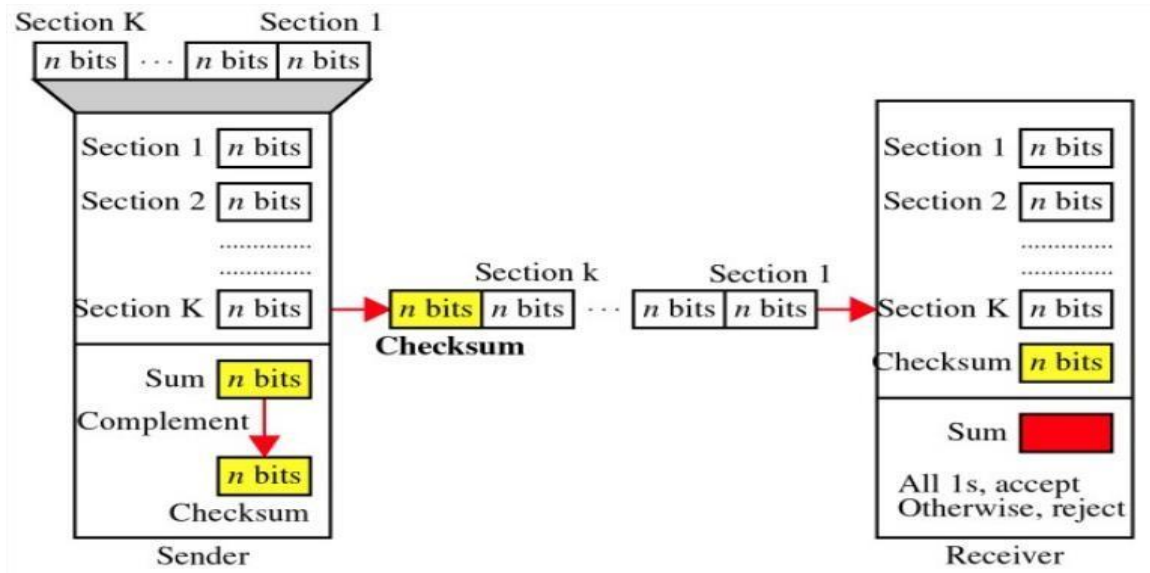
The error detection method used by the higher layer protocols. Like other methods, it depends on the concept of redundancy. In the sender, the checksum generator subdivided the data into equal segments of bits. These segments are added together using one's complement arithmetic in such a way that the total is also n bits long. That total is then complemented and appended to the end of the original data unit as redundancy bits called the checksum field. The data segment is referred to as T and the checksum as $-T$.

To create the checksum the sender does the following:

- The unit is divided into k sections, each of n bits.
- Sections 1 and 2 are added together using one's complement.
- Section 3 is added to the result of the previous step.

- Section 4 is added to the result of the previous step.
- The Process repeats until section k is added to the result of the previous step.
- The final result is complemented to make the checksum.

In the receiver, the coming data is subdivided into data and checksum. The total value of the data unit and the checksum must be zero for errorless transmission.



Example

Suppose the following block of 16 bits is to be sent using a checksum of 8 bits.

10101001 00111001

The numbers are added using one's complement

	10101001
	00111001

Sum	11100010
Checksum	00011101

The pattern sent is 10101001 00111001 00011101

Now suppose the receiver receives the pattern sent and there is no error.

10101001 00111001 00011101

When the receiver adds the three sections, it will get all 1s, which, after complementing, is all 0s and shows that there is no error.

	10101001	
	00111001	
	00011101	
Sum	11111111	
Complement	00000000	means that the pattern is OK.

Now suppose there is a burst error of length 5 that affects 4 bits.

10101111 1111001 00011101

When the receiver adds the three sections, it gets

	10101111	
	11111001	
	00011101	
Partial Sum	1 11000101	
Carry		1
Sum	11000110	
Complement	00111001	the pattern is corrupted.



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Computer Network Fundamentals

Chapter Five

IP Addressing **((Network Layer))**

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5.1- Introduction

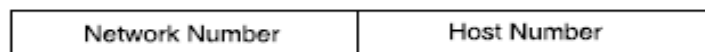
Communication at the network layer is source to destination; a computer somewhere in the world needs to communicate with another computer somewhere else in the world. Usually, computers communicate through the Internet. The packet transmitted by the sending computer may pass through several LANs or WANs before reaching the destination computer.

For this level of communication, we need a **global addressing scheme called logical addressing**. Today, we use the term IP address to mean a logical address in the network layer of the TCP/IP protocol suite.

5.2- What is an IP address?

In the most widely installed level of the Internet Protocol (IP) today, an IP address is a **32-bit number** that *uniquely* and *universally* defines the connection of a device (for example, a computer or a router) to the Internet. The 32 bits in length gives us a maximum of 2^{32} addresses. These addresses are referred to as **IPv4** (IP version 4) addresses or simply IP addresses if there is no confusion.

An IP address has *two* parts: The first part of an Internet address identifies the network (NetID) on which the host resides, while the second part identifies the particular host on the given network (HostID). This creates the two-level addressing hierarchy as shown in figure below

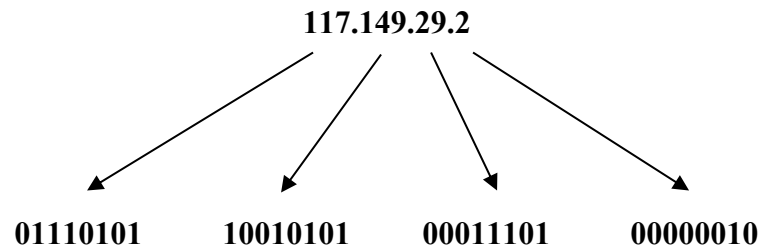


Each device must know its own IP address and the IP address of the device with which it needs to communicate (source and destination). To communicate on the Internet, every organization must have at least one valid Internet IP address. The 32-bits IP address is known as IPv4.

5.2.1- Notations

There are two prevalent notations to show an IPv4 address: binary notation and dotted decimal notation.

Figure below shows an IPv4 address in both binary and dotted-decimal notation. Note that because each byte (octet) is 8 bits, each number in dotted-decimal notation is a value ranging from 0 to 255.



IP addressing can be classified as **classful** and **classless** addresses.

5.3- Classful Addressing

IPv4 addressing, at its inception, used the concept of classes. This architecture is called classful addressing. Although this scheme is becoming obsolete, we briefly discuss it here to show the rationale behind classless addressing.

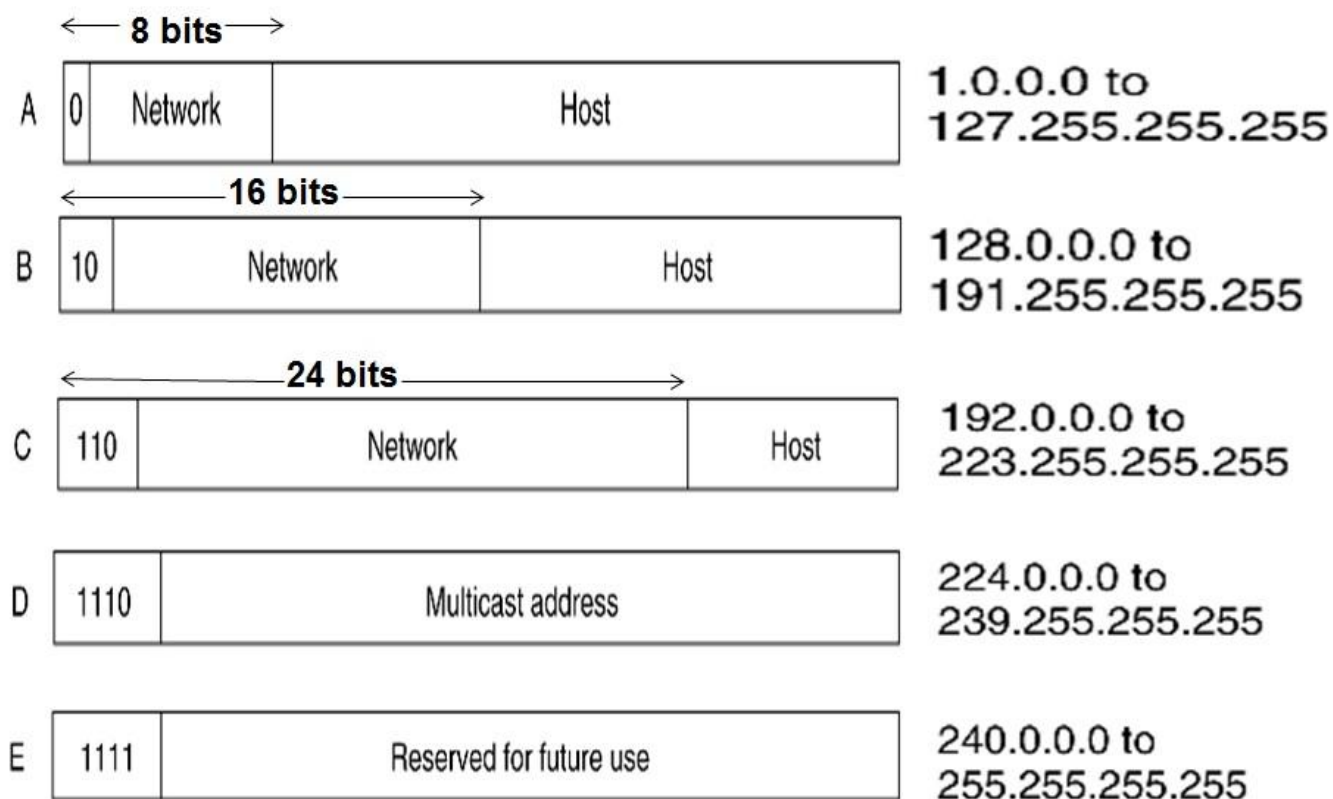
5.3.1- Classes

In classful addressing, the address space is divided into **five classes: A, B, C, D, and E**. Each class occupies some part of the address space.

In classful addressing, an IP address in class A, B, or C is divided into *netid* and *hostid*. These parts are of varying lengths, depending on the class of the address. Figure below shows some *netid* and *hostid* bytes. Note that the concept does not apply to classes D and E.

In class A, one byte defines the netid and three bytes define the hostid. In class B, two bytes define the netid and two bytes define the hostid. In class C, three bytes define the netid and one byte defines the hostid.

Class D addresses were designed for multicasting. Each address in this class is used to define one group of hosts on the Internet. And lastly, the class E addresses were reserved for future use.



Class	High Order Bits	First Octet Range	Number of Network Bits	Number of Host Bits	Number of Networks	Number of Hosts per Network
A	0	0-127	8	24	128	16,777,216
B	10	128-191	16	16	16,384	65,536
C	110	192-223	24	8	2,097,152	256
D	1110	224-239	Used for Multicasting to multiple hosts.			
E	1111	240-255	Reserved for research and development.			

5.3.2- Reserved IP Addresses

✓ **Network address and broadcast address**

For each class there are two addresses which are reserved, these are the first and last address. The first address identifies the *NetID address*, while the last address in the class range is the *broadcast address*, as shown in table below

Class	NetID address	Valid IP addresses	Broadcast address
A	125.0.0.0	125.0.0.1 – 125.255.255.254	125.255.255.255
B	180.20.0.0	180.20.0.1 – 180.20.255.254	180.20.255.255
C	220.100.10.0	220.100.10.1 – 220.100.10.254	220.100.10.255

✓ **Loopback address**

The address **127.0.0.1** is also reserved which is known as a *loopback address* which is used to allow a node to send a test packet to itself without generating a network traffic for the purpose of testing the TCP/IP configuration on a PC.

✓ **Link Local Addresses:**

Address Range (**169.254.0.0 to 169.254.255.255**), these addresses are automatically assigned by the operating system when there is no IP configuration is available and the DHCP server not available.

5.3.3- Private IP addresses

These addresses can be used on a private network, but they are not routable through the internet. The range for the private IP addresses are given in table below

Range	Prefix
10.0.0.0 – 10.255.255.255	/ 8
172.16.0.0 – 172.31.255.255	/ 12
192.168.0.0 – 192.168.255.255	/ 16

5.3.4- Subnet Mask

Although the length of the netid and hostid (in bits) is predetermined in classful addressing, we can also use a mask (also called the default mask), a 32-bit number made of contiguous 1's followed by contiguous 0's. The masks for classes A, B, and C are shown in Table below; The concept does not apply to classes D and E.

The mask can help us to find the network address (netid) by making a logical AND operation between the IP address and its corresponding mask.

Class	Binary	Dotted Decimal	Prefix
A	11111111 00000000 00000000 00000000	255.0.0.0	/ 8
B	11111111 11111111 00000000 00000000	255.255.0.0	/ 16
C	11111111 11111111 11111111 00000000	255.255.255.0	/ 24

The last column in the table shows the mask in the form (/n) where n can be 8, 16, or 24 in classful addressing. This notation is also called **prefix notation** or Classless Interdomain Routing (CIDR) notation. The notation is used in classless addressing. We introduce it here because it can also be applied to classful addressing. We will show later that classful addressing is a special case of classless addressing.

5.4- Classless Addressing

Classful allocation of address space often wasted many addresses, which exhausted the availability of IPv4 addresses. For example, a company that had a network with 260 hosts would need to be given a class B address with more than 65,000 addresses. The system that we currently use is referred to as classless addressing. With the classless system, address blocks appropriate to the number of hosts are assigned to companies or organizations without regard to the unicast class. This is can be achieved using many techniques such as: Subnetting, Variable Length Subnet Mask (VLSM), and Network Address Translation (NAT)

5.5- Subnetting

For the purpose of *reducing network traffics* and *simplifying network management*, and also *reducing the waste in IP addresses*; large network can be divided into smaller networks (subnetworks). Subnetting allows for creating multiple logical networks from a single address block.

We create the subnets by using one or more of the host bits as network bits. *This is done by extending the mask to borrow some of the bits from the host portion of the address to create additional network bits.* The more host bits used, the more subnets that can be defined. For each bit borrowed, we double the number of subnetworks available.

Notes:

- *The maximum number of borrowed bits is by leaving 2-bits in the HostID part of the address.*
- To determine the number of subnets or hosts use the following rules:

$$\text{Number of Subnets} = 2^{\text{number_of_borrowed_host_bits}}$$

$$\text{Usable Hosts Per Subnet} = (2^{\text{number_of_remaining_host_bits}} - 2)$$

Example: What are the number of subnets, hosts, and the valid host addresses given class C IP address 192.168.10.5 and a subnet mask 255.255.255.192 (192.168.10.5 / 26)

Solution:

IP address → 192.168.10.5 → 11000000 . 10101000 . 00001010 . 00000101

Subnet mask → 255.255.255.192 → 11111111 . 11111111 . 11111111 . 11000000

Network address → 192.168.10.0 → 11000000 . 10101000 . 00001010 . 00000000

Subnet mask $\rightarrow 255.255.255.192 \rightarrow 11111111.11111111.11111111.\overline{11}000000$

2 bits are borrowed

The number of subnets = 2^n . (where n = the number of bits borrowed)

$$= 2^2 = 4 \text{ subnets}$$

The number of hosts on each subnet = $2^m - 2$. (where m = the number of bits left for hosts).

$$= 2^6 - 2 = 62$$

shows that each of these subnets can have 62 hosts.

For each subnet, examine the last octet in binary. The values in these octets for the four networks are:

Subnet 1: **00000000** = 0

Subnet 2: **01000000** = 64

Subnet 3: **10000000** = 128

Subnet 4: **11000000** = 192

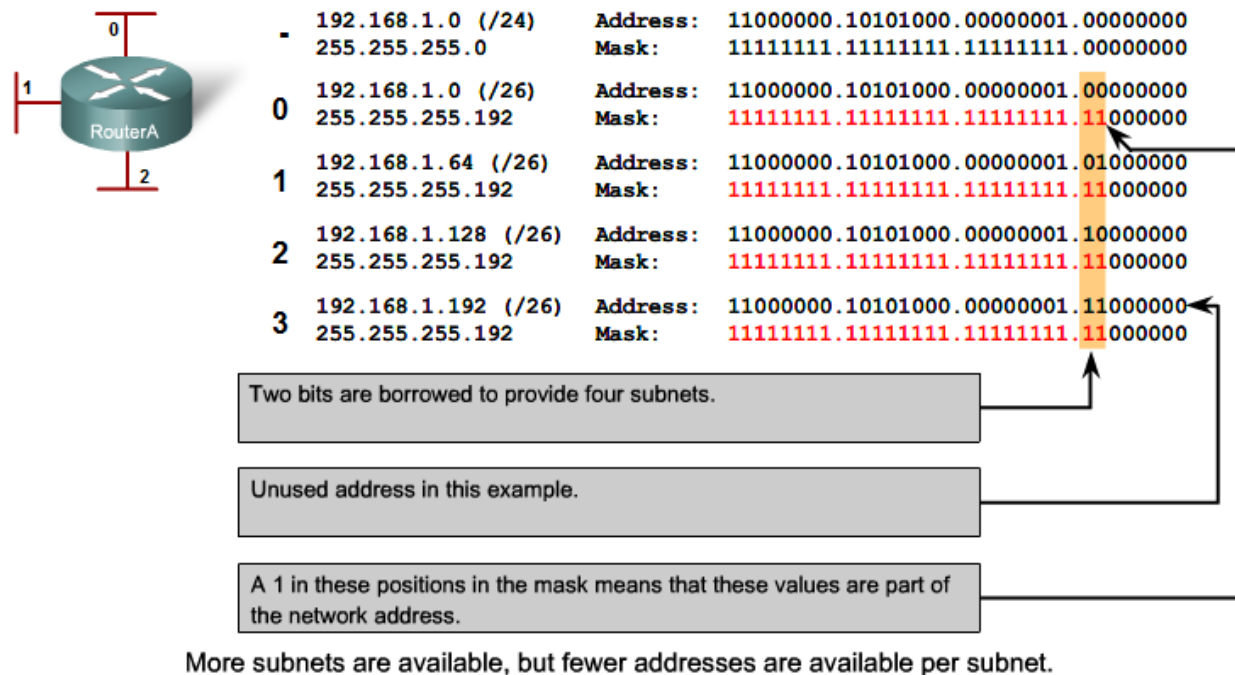
Or the subnet block size is = $2^{\text{number-of-host-bits}} = 2^6 = 64$ (*the magic number*)

Therefore, the subnets are = (0, 64, 128, 192)

Table below shows the valid host addresses and the subnet and broadcast addresses for each subnet.

Subnet	Subnet address	Host range	Broadcast address
1	192.168.10.0	192.168.10. 1 $\rightarrow$ 192.168.10. 62	192.168.10.63
2	192.168.10.64	192.168.10. 65 $\rightarrow$ 192.168.10. 126	192.168.10.127
3	192.168.10.128	192.168.10. 129 $\rightarrow$ 192.168.10. 190	192.168.10.191
4	192.168.10.192	192.168.10. 193 $\rightarrow$ 192.168.10. 254	192.168.10.255

Figure below illustrate the same process



A serious limitation of using only a single subnet mask across a given network was that an organization is locked into a fixed-number of fixed-sized subnets. The solution to this problem is to use a VLSM (variable length subnet mask) enables a network number to be configured with different subnet masks on different interfaces.

5.6- IPv6 Addresses

The need for more addresses, in addition to other concerns about the IP layer, motivated a new design of the IP layer called the new generation of IP or IPv6 (IP version 6). In this version, the Internet uses **128-bit addresses** that give much greater flexibility in address allocation. These addresses are referred to as IPv6 (IP version 6) addresses.

An IPv6 address consists of 16 bytes (octets); it is 128 bits long. To make addresses more readable, IPv6 specifies hexadecimal colon notation. In this notation, 128 bits is divided into eight sections, each 2 bytes in length. Two bytes in hexadecimal notation requires four hexadecimal digits. Therefore, the address consists of 32 hexadecimal digits, with every four digits separated by a colon, as shown below

FDEC: 0074 : 0000 : 0000 : 0000 : BOFF : 0000 : FFF0